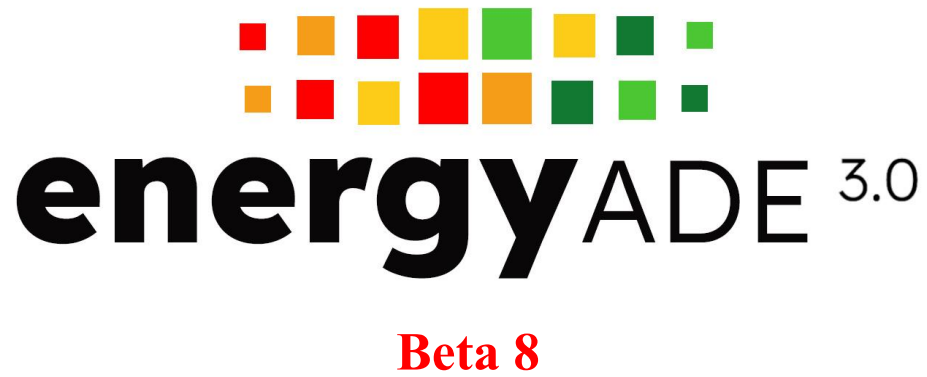




**DRAFT**

Last updated on 2 February 2026





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The latest version of this document can be downloaded from:  
[https://github.com/tudelft3d/Energy\\_ADE/tree/3.0beta8/documentation](https://github.com/tudelft3d/Energy_ADE/tree/3.0beta8/documentation)

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## **Part 1**

# **Overview of the changes since **beta 7****

## Introduction

The CityGML Energy ADE 3.0 **beta 8** builds upon the previous version and adds a number of changes resulting mainly from the interviews and discussion between the author of this document and several experts (some of them are listed in the Acknowledgements, see colophon) during summer and autumn 2025.

At the moment, this is the only documentation available and it consists of UML diagrams, a detailed changelog, and the visual mapping between the conceptual data model and the database implementation for the 3D City Database version 4.x.

The changes of the current version compared to the previous one can be grouped into few major areas. They are:

1) Better alignment with the [Energy ADE 2.0](#) (released in 2019), which contains some improvements over the Energy ADE 1.0 despite the interruption of its development in 2019. The changes in this beta 8 regard mainly the **Layered Construction module** and the addition of the new **MonthlyTimeSeries** (and related classes) in the **Time series module**. For the former, class **AbstractMaterial** and its subclasses have been enriched with some further attributes that allow for the modelling of multi-glazing windows. In the case of the Time series module, new classes for monthly values have been added, as they represent indeed a special, but relevant case of *irregular* time series. Finally, several change requests listed in the (now dormant) GitLab repository of the Energy ADE 2.0 have been checked and – wherever reasonable – integrated into the current version. The full list is still available here: <https://git.rwth-aachen.de/energyade/citygml-energy/-/issues>

2) Better alignment with the CityGML 3.0 data model. An overall redesign of the Energy ADE classes derived from **gml:AbstractFeature** (but that are not derived from class **core:\_CityObject**) has allowed to introduce the concepts of (permanent) *identifier* and *temporal validity extents*. The same goes for the classes derived from **\_CityObject**, thanks to the injection of such properties via the ADE-hook mechanism. In CityGML 3.0, such attributes are found in classes **AbstractFeature** and **AbstractFeatureWithLifeSpan**. Finally, all ADE feature classes have now the possibility to have *external references*, just like classes derived from **\_CityObject**. This is particularly useful in case of feature-based classes like **EnergyPerformanceCertificate**. Beside backporting some useful concepts from CityGML 3.0, this further alignment will facilitate the future porting of the Energy ADE 3.0 to CityGML 3.0.

3) Following the observations and suggestions of the interviewed people, the former **Building Physics** and **Occupancy modules** have been reworked and merged into the new **Building module**. In short, the previous dichotomy of classes **UsageZone** (with occupants) and **ThermalZone** (with physical properties) has been resolved by "fusing" such classes into a new common class named **Zone**. Additionally, this modelling approach is more in line with what happens with gbXML, EnergyPlus or CitySim, for example.

4) Following the comments and suggestions from the previous beta 7, the overall modelling approach of the (former) thermal boundaries of a (former) thermal zone has been improved. A **Zone** is now bounded by *zone thematic surfaces*, which are clearly distinct from CityGML "normal" thematic surfaces in order to avoid using the same classes for multiple, distinct objects (e.g. a LoD2 building and a LoD2 zone). Additionally, the **PartyWallSurface** class has been dropped and replaced by a simple Boolean attribute *isShared* inherited by the **\_BoundarySurface** class via ADE-

hook. This extends the possibility to have not only shared walls, but any other shared thematic surfaces, be it for buildings or zones.

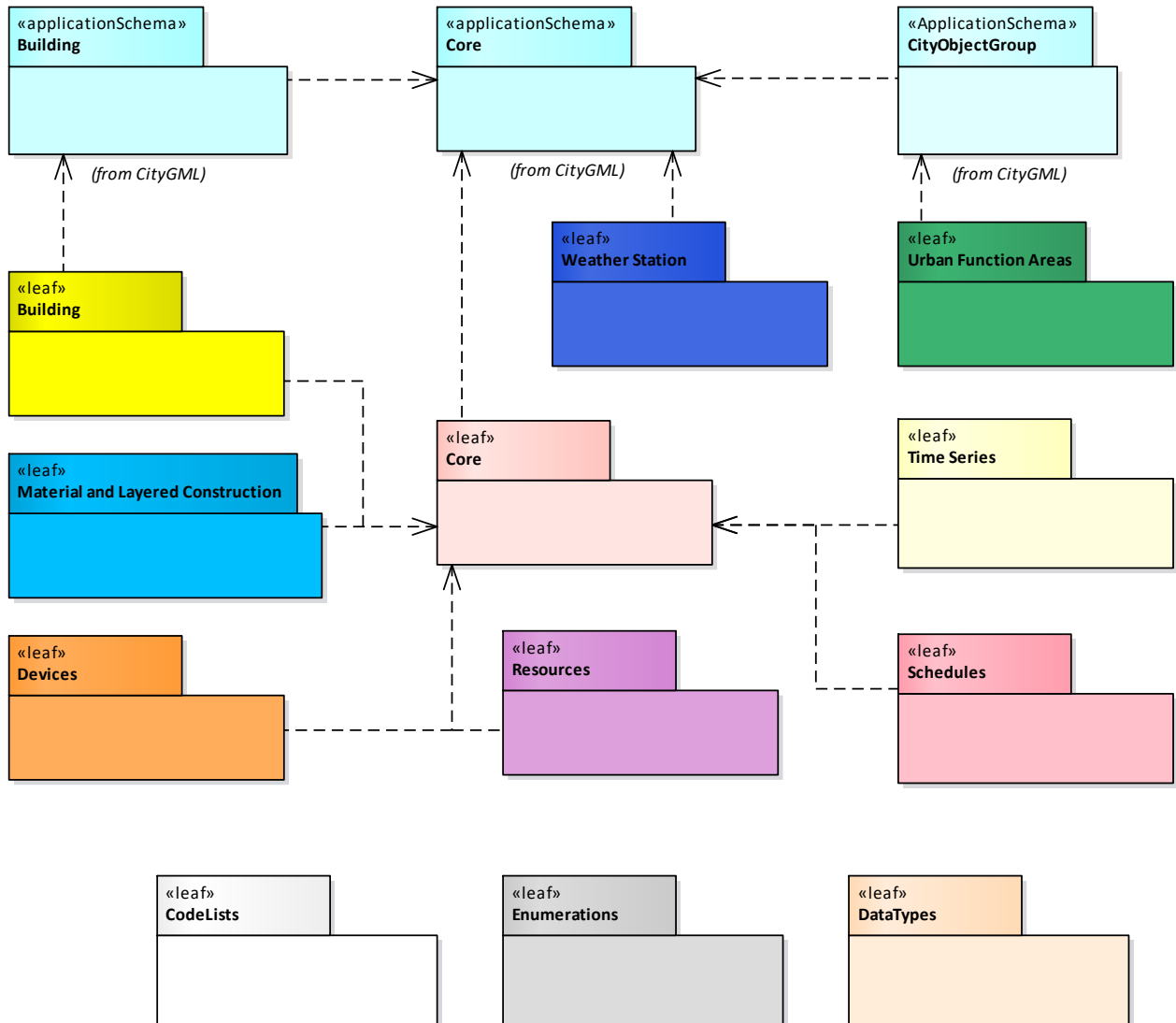
- 5) A detailed list of changes can be found in the Changelog section of this document
- 6) From the UML diagram, the XSD file of the Energy ADE 3.0 beta 8 has been derived using ShapeChange 3.10. The resulting XSD file has been additionally "cleaned" and checked manually.
- 7) Starting from the XSD file, the DDL scripts for the Energy ADE database schema for the 3D City Database 4.x have been derived. Also in this case, some additional manual customization has been carried out to harmonise the name of tables, columns, etc. and to reduce the overall number of otherwise automatically generated tables.
- 8) Test data are currently being cleaned up and prepared. When ready, they will be available on the GitHub repository. The dataset "Alderaan\_Energy\_ADE.gml" is expected to contain nearly all classes of the Energy ADE 3.0. It will be possible to visualise it and explore the attributes using the [KIT Modelviewer 7.4](#) (or higher) after copying the XSD file of the Energy ADE in folder \GMLSchemata\CityGML\_2\_0\CityGML

You are heartily invited to have a look at the current data model and to leave comments and suggestions in GitHub. You can open an issue regarding the **3.0beta8** development branch, following this link: [https://github.com/tudelft3d/Energy\\_ADE/issues/4](https://github.com/tudelft3d/Energy_ADE/issues/4)

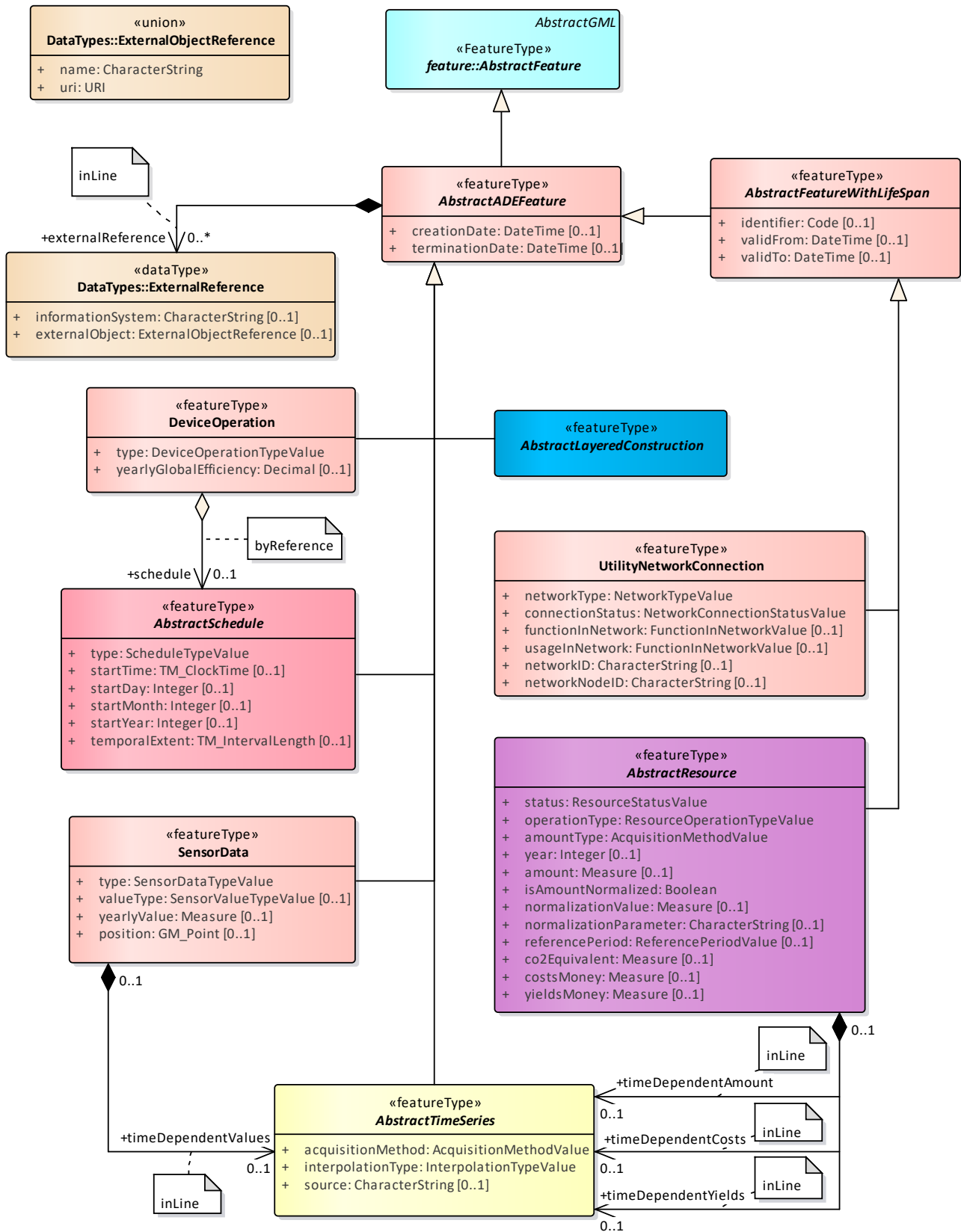
# **Part 2**

## **UML diagrams**

## Overview of the packages



## Core module: *AbstractADEFeature* and related classes





## Core module: Codelists and Enumerations

| «codeList»<br>CodeLists::<br>AcquisitionMethodValue                                  |
|--------------------------------------------------------------------------------------|
| + measurement<br>+ simulation<br>+ calibratedSimulation<br>+ estimation<br>+ unknown |

| «codeList»<br>CodeLists::<br>DeviceOperationTypeValue                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + cooking<br>+ domesticHotWater<br>+ electricalAppliances<br>+ lighting<br>+ otherOrCombination<br>+ process<br>+ shading<br>+ spaceCooling<br>+ spaceHeating<br>+ ventilation<br>+ unknown |

| «codeList»<br>CodeLists::<br>FunctionInNetworkValue             |
|-----------------------------------------------------------------|
| + producing<br>+ consuming<br>+ storage<br>+ other<br>+ unknown |

| «codeList»<br>CodeLists::<br>ScheduleTypeValue                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + hour<br>+ day<br>+ week<br>+ month<br>+ season<br>+ year<br>+ typicalDay<br>+ typicalWeek<br>+ typicalMonth<br>+ typicalSeason<br>+ typicalYear<br>+ monday<br>+ tuesday<br>+ wednesday<br>+ thursday<br>+ friday<br>+ saturday<br>+ sunday<br>+ weekday<br>+ weekendDay<br>+ beerDay<br>+ holiday<br>+ january<br>+ february<br>+ march<br>+ april<br>+ may<br>+ june<br>+ july<br>+ august<br>+ september<br>+ october<br>+ november<br>+ december<br>+ spring<br>+ summer<br>+ autumn<br>+ winter<br>+ other |

| «codeList»<br>CodeLists::<br>NetworkTypeValue                                                                         |
|-----------------------------------------------------------------------------------------------------------------------|
| + data<br>+ gas<br>+ districtHeating<br>+ districtCooling<br>+ wasteWater<br>+ electricalGrid<br>+ other<br>+ unknown |

| «codeList»<br>CodeLists::<br>RelationTypeValue |
|------------------------------------------------|
| + adjacentTo<br>+ installedOn<br>+ sharedWith  |

| «codeList»<br>CodeLists::<br>ResourceOperationTypeValue        |
|----------------------------------------------------------------|
| + demands<br>+ has<br>+ hasBalanceOf<br>+ produces<br>+ stores |

| «codeList»<br>CodeLists::WeatherDataValue                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + airTemperature<br>+ humidity<br>+ windSpeed<br>+ windDirection<br>+ cloudiness<br>+ globalSolarIrradiance<br>+ directSolarIrradiance<br>+ diffuseSolarIrradiance<br>+ terrestrialEmission<br>+ downwardTerrestrialRadiation<br>+ daylightIlluminance<br>+ shadowingFactor<br>+ precipitation |

| «codeList»<br>CodeLists::<br>SensorValueTypeValue                                                            |
|--------------------------------------------------------------------------------------------------------------|
| + absolute maximum<br>+ average maximum<br>+ average<br>+ average minimum<br>+ absolute minimum<br>+ unknown |

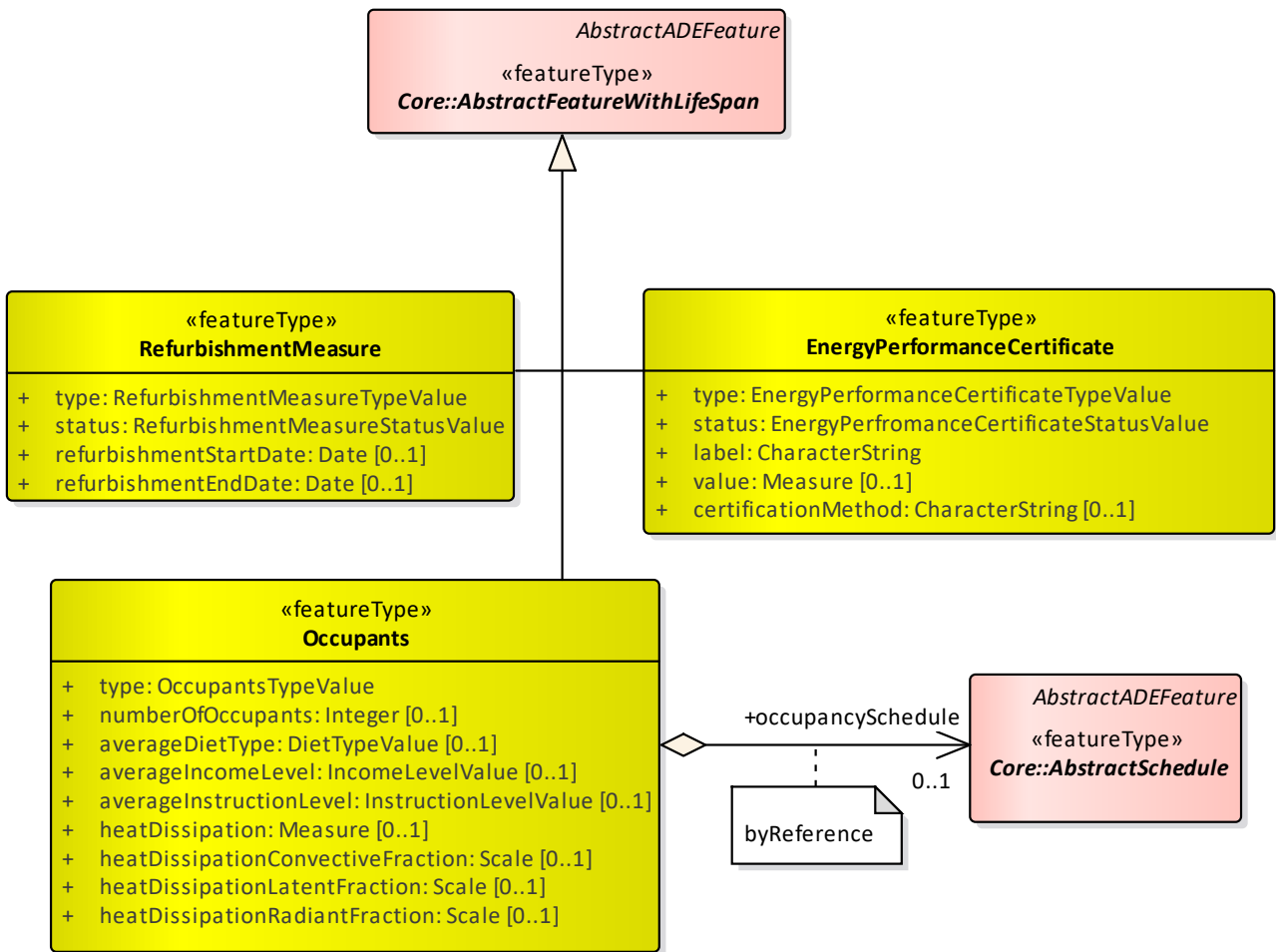
| «codeList»<br>CodeLists::<br>ReferencePeriodValue                                                      |
|--------------------------------------------------------------------------------------------------------|
| + minute<br>+ hour<br>+ day<br>+ week<br>+ month<br>+ hot season<br>+ cold season<br>+ year<br>+ other |

| «enumeration»<br>Enumerations::<br>InterpolationTypeValue                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| averageInPrecedingInterval<br>averageInSucceedingInterval<br>constantInPrecedingInterval<br>constantInSucceedingInterval<br>continuous<br>discontinuous<br>instantaneousTotal<br>maximumInPrecedingInterval<br>maximumInSucceedingInterval<br>minimumInPrecedingInterval<br>minimumInSucceedingInterval<br>precedingTotal<br>succeedingTotal |

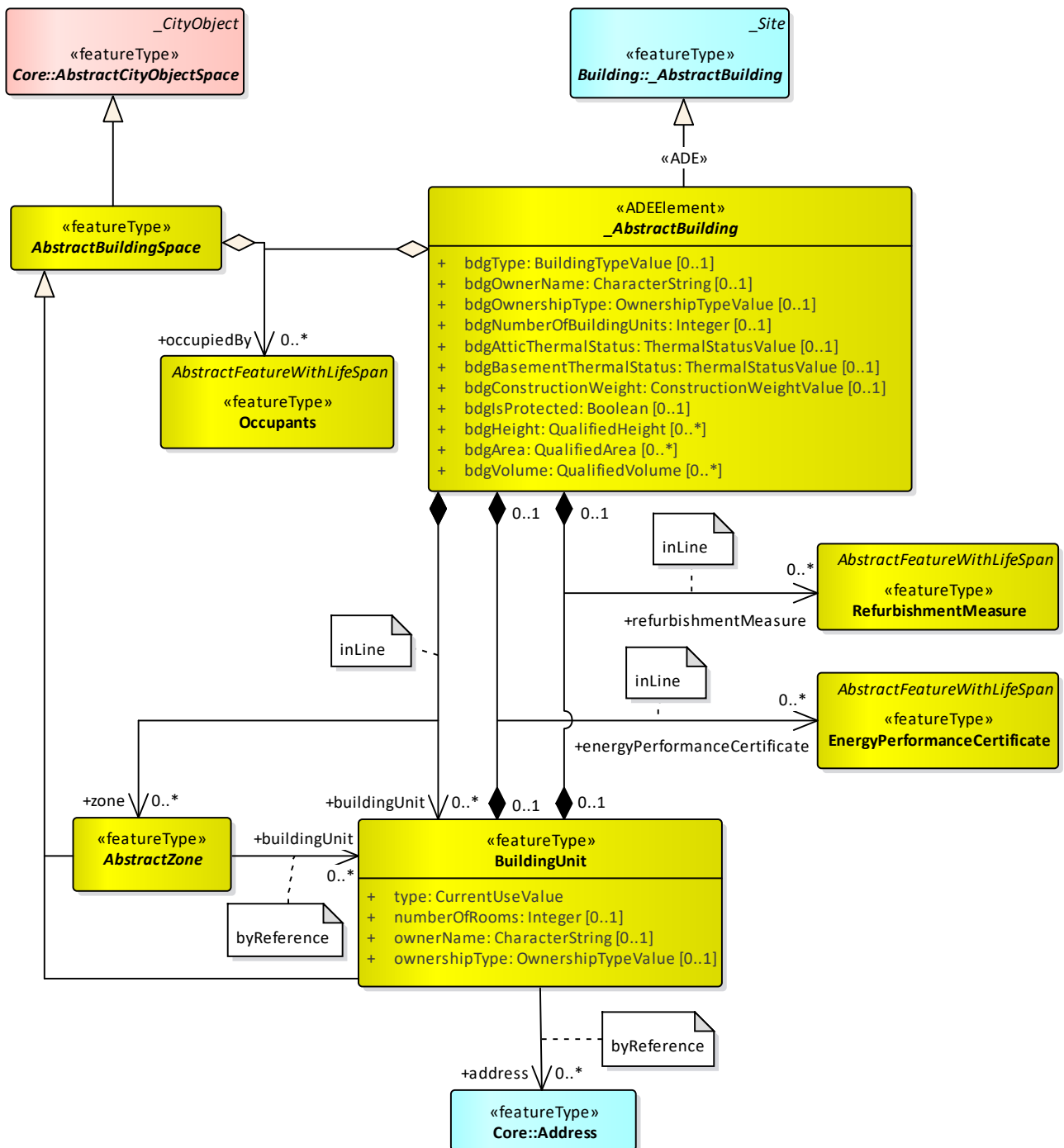
| «enumeration»<br>Enumerations::<br>NetworkConnectionStatusValue                     |
|-------------------------------------------------------------------------------------|
| connected<br>disconnectedButConnectable<br>disconnectedAndNotConnectable<br>unknown |

| «enumeration»<br>Enumerations::<br>ResourceStatusValue |
|--------------------------------------------------------|
| actual<br>potential<br>unknown                         |

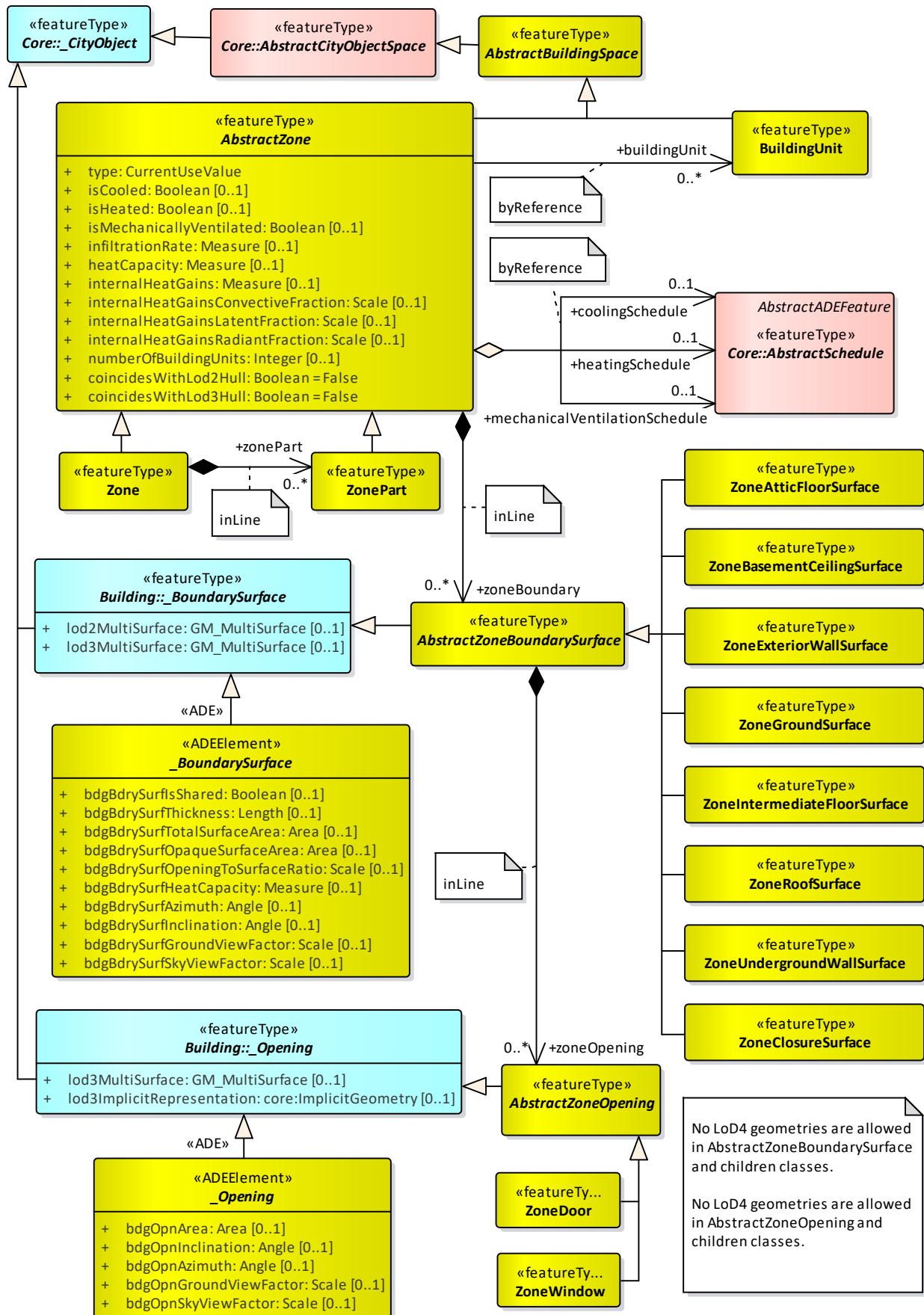
## Building module: Features



## Building module: ADE *AbstractBuilding* and related classes



## Building module: *AbstractZone* and related classes



## Building module: Codelists and Enumerations

| «codeList»<br>CodeLists::<br>BuildingTypeValue                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + single-family house<br>+ terraced house<br>+ multi-family house<br>+ apartment block<br>+ commercial building<br>+ industrial building<br>+ office building<br>+ religious building<br>+ hospital<br>+ leisure activity building<br>+ unknown |

| «codeList»<br>CodeLists::<br>ConstructionWeightValue       |
|------------------------------------------------------------|
| + veryLight<br>+ light<br>+ medium<br>+ heavy<br>+ unknown |

| «codeList»<br>CodeLists::OwnershipTypeValue                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + corporation<br>+ government<br>+ nonOccupantPrivateOwner<br>+ nonProfitOrganisation<br>+ occupantPrivateOwner<br>+ otherOrCombination<br>+ propertyManagementCompany<br>+ unknown |

| «codeList»<br>CodeLists::<br>RefurbishmentMeasureTypeValue                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + installedSolarPanels<br>+ installedPVPanels<br>+ installedHeatPump<br>+ changedBoiler<br>+ insulatedWholeThermalHull<br>+ installedFloorHeating<br>+ changedWindows<br>+ changedDoors<br>+ insulatedRoof<br>+ insulatedExteriorWalls<br>+ insulatedGroundFloor<br>+ other<br>+ unknown |

| «codeList»<br>CodeLists::<br>OccupantsTypeValue                                                         |
|---------------------------------------------------------------------------------------------------------|
| + patients<br>+ residents<br>+ students<br>+ visitors<br>+ workers<br>+ otherOrCombination<br>+ unknown |

| «codeList»<br>CodeLists::IncomeLevelValue                             |
|-----------------------------------------------------------------------|
| + very low<br>+ low<br>+ middle<br>+ high<br>+ very high<br>+ unknown |

| «codeList»<br>CodeLists::<br>RefurbishmentMeasureStatusValue        |
|---------------------------------------------------------------------|
| + completed<br>+ planned<br>+ suggested<br>+ cancelled<br>+ unknown |

| «codeList»<br>CodeLists::CurrentUseValue |
|------------------------------------------|
|------------------------------------------|

| «codeList»<br>CodeLists::InstructionLevelValue                                                                 |
|----------------------------------------------------------------------------------------------------------------|
| + no formal education<br>+ primary school<br>+ secondary school<br>+ university<br>+ postGraduate<br>+ unknown |

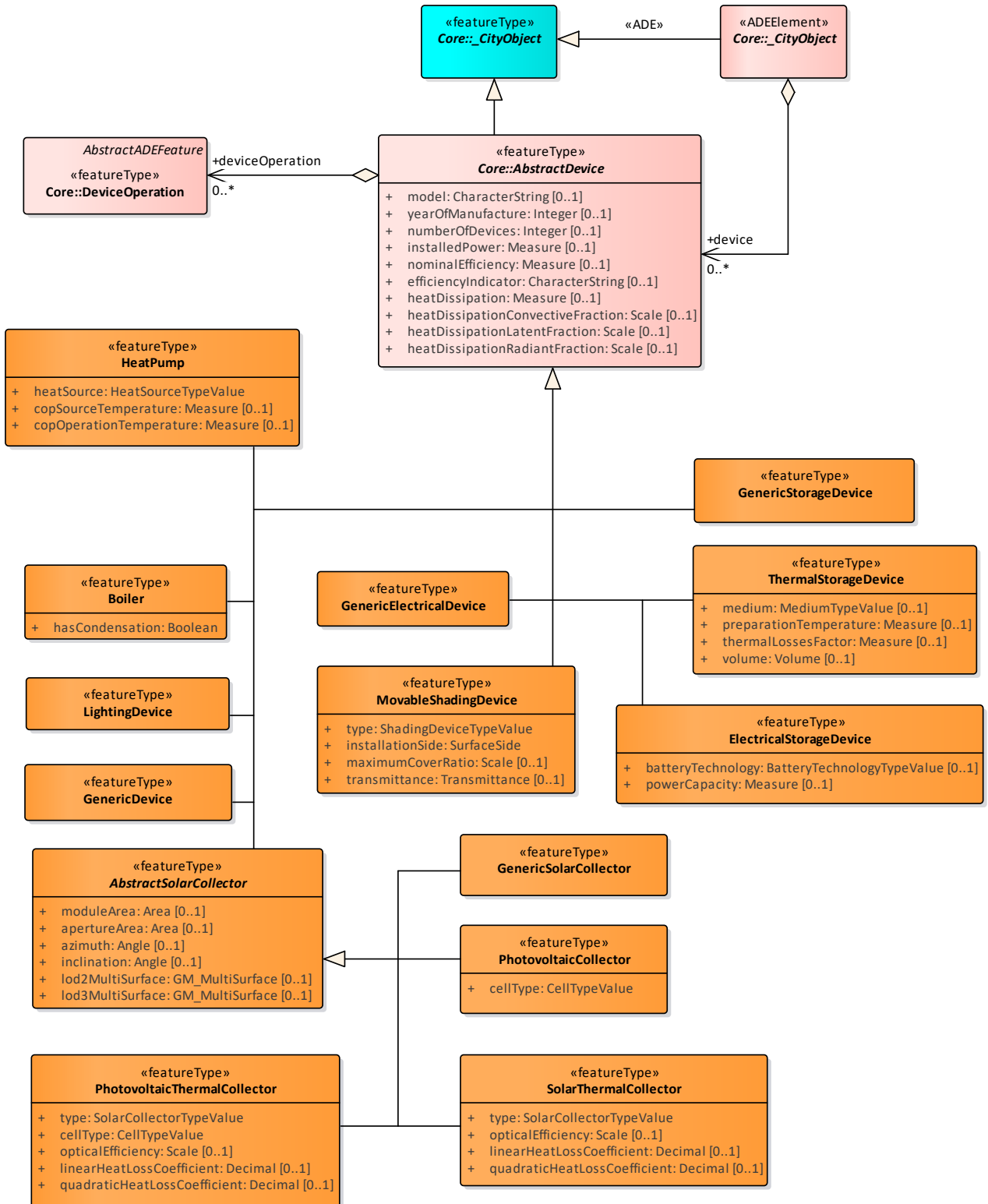
| «codeList»<br>CodeLists::<br>EnergyPerformanceCertificateTypeValue |
|--------------------------------------------------------------------|
|--------------------------------------------------------------------|

| «enumeration»<br>Enumerations::ThermalStatusValue                                        |
|------------------------------------------------------------------------------------------|
| isOnlyHeated<br>isOnlyCooled<br>isHeatedAndCooled<br>isNeitherHeatedNorCooled<br>unknown |

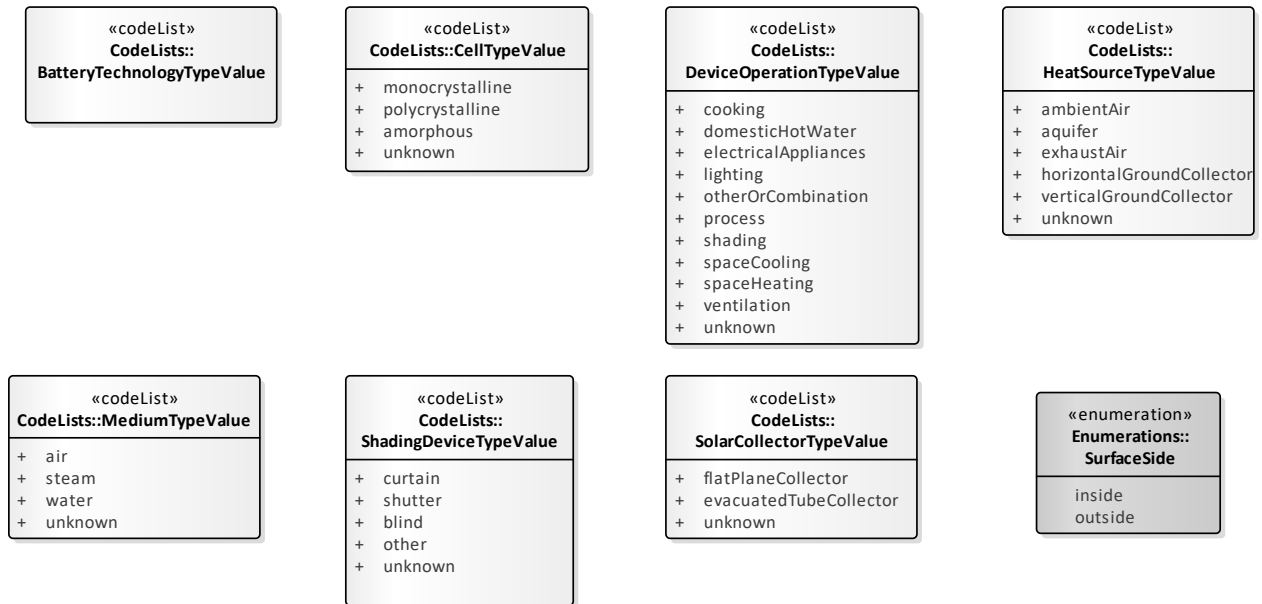
| «codeList»<br>CodeLists::DietTypeValue                          |
|-----------------------------------------------------------------|
| + omnivorous<br>+ vegetarian<br>+ vegan<br>+ other<br>+ unknown |

| «codeList»<br>CodeLists::<br>EnergyPerformanceCertificateStatusValue |
|----------------------------------------------------------------------|
| + measured<br>+ estimated<br>+ simulated<br>+ unknown                |

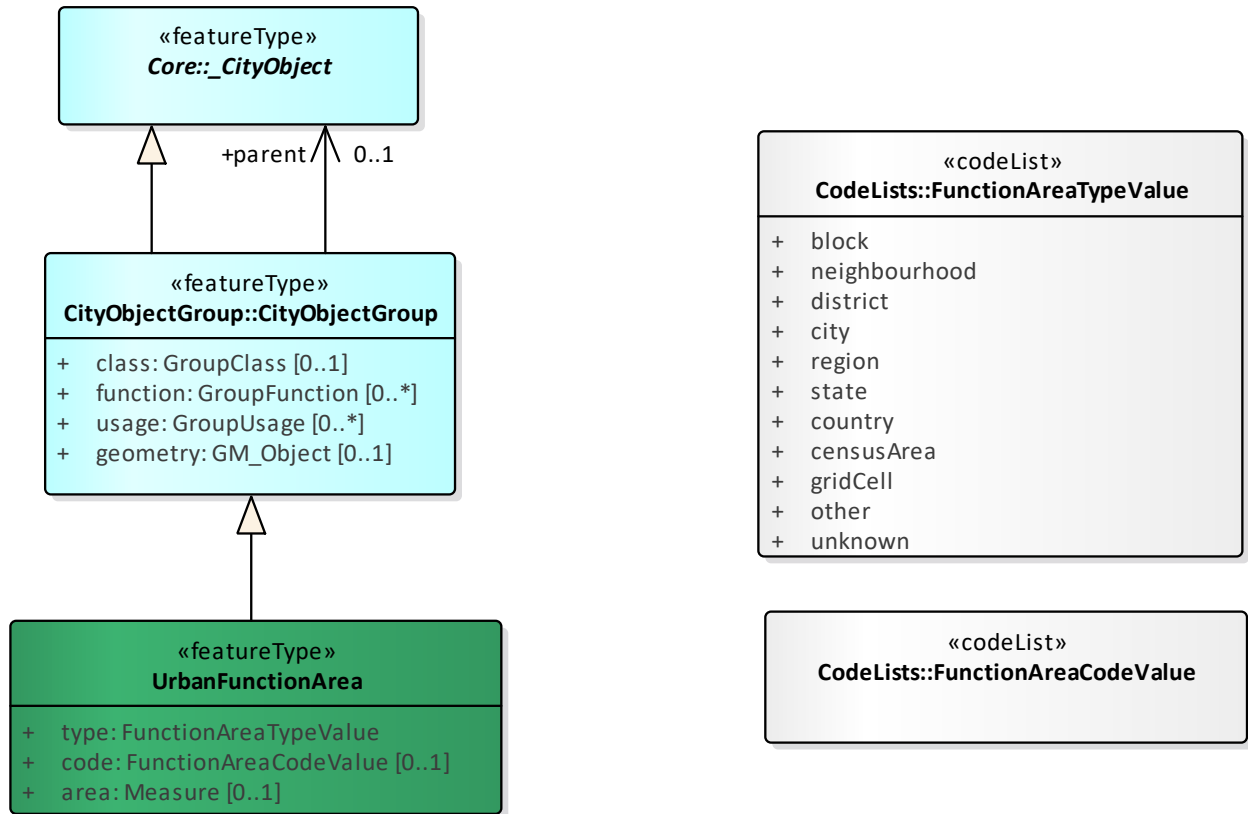
## Devices module



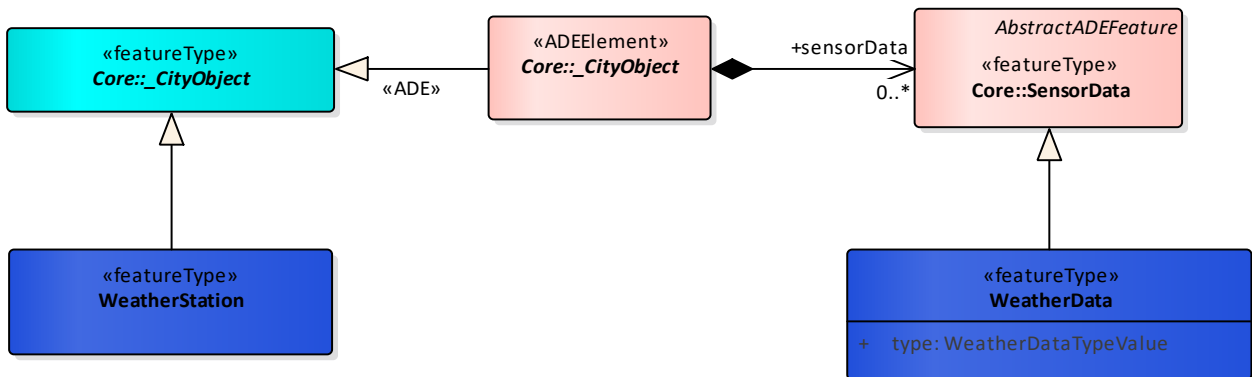
## Devices module: Codelists and Enumerations



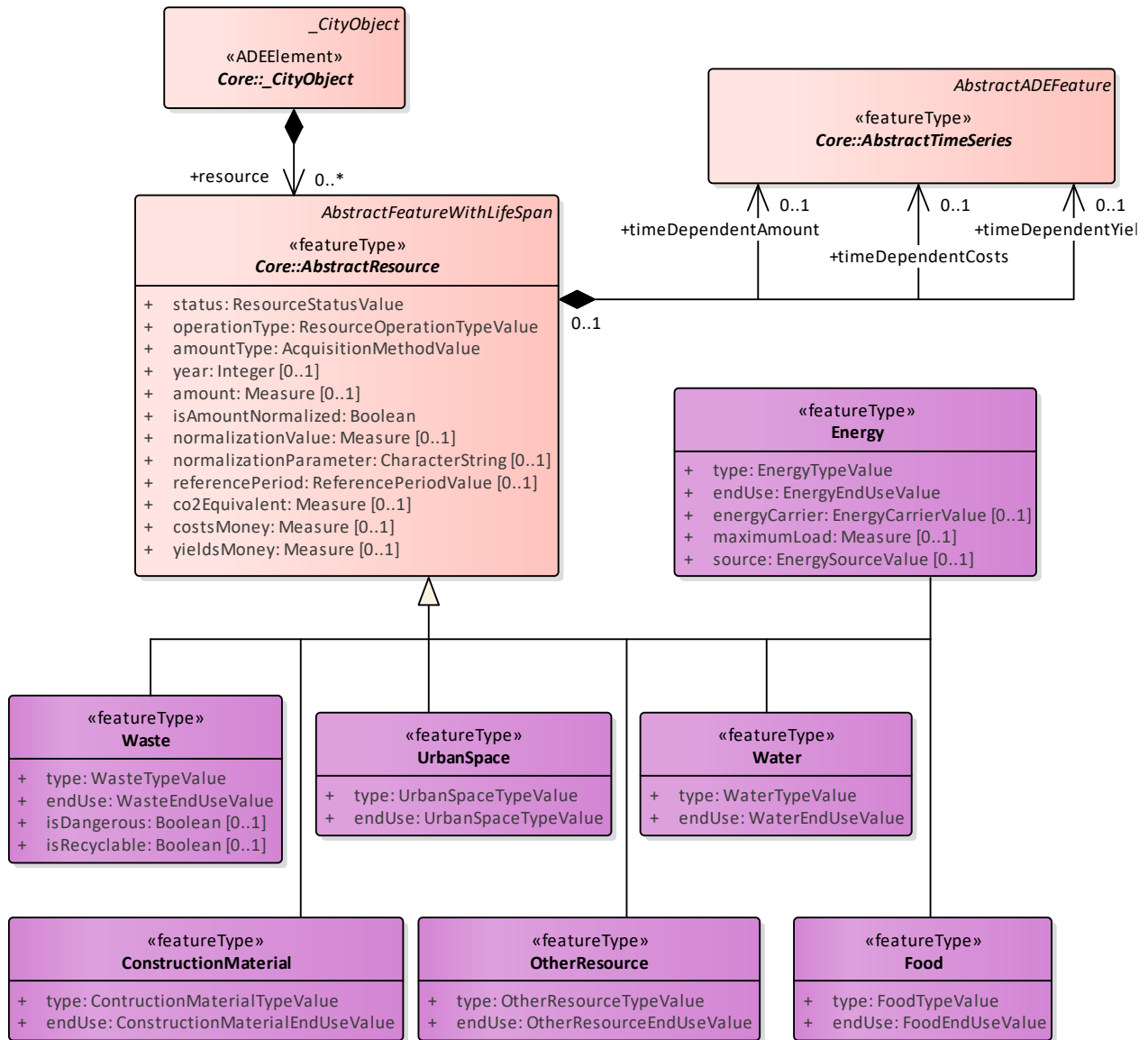
## Urban function areas module



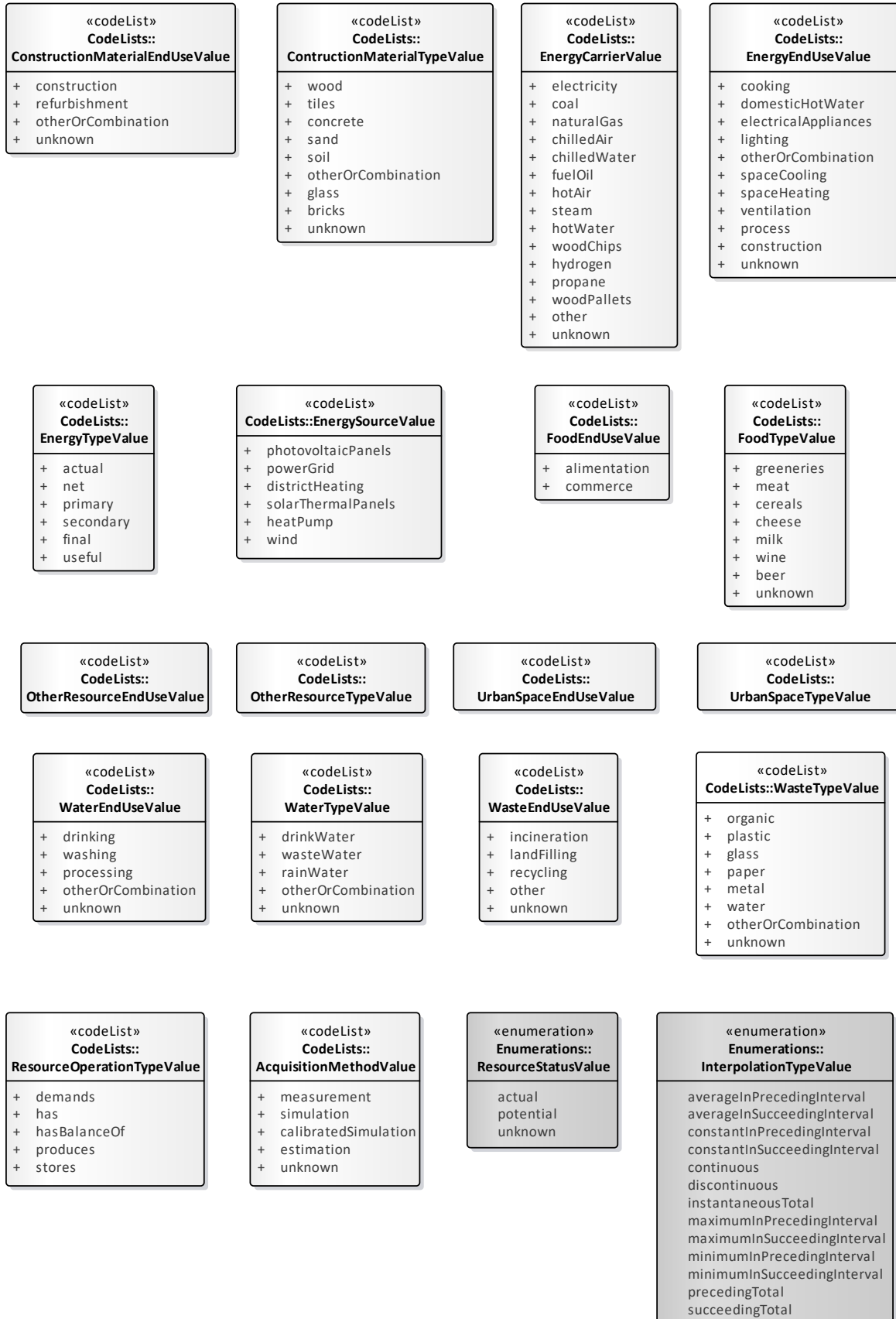
## Weather station module

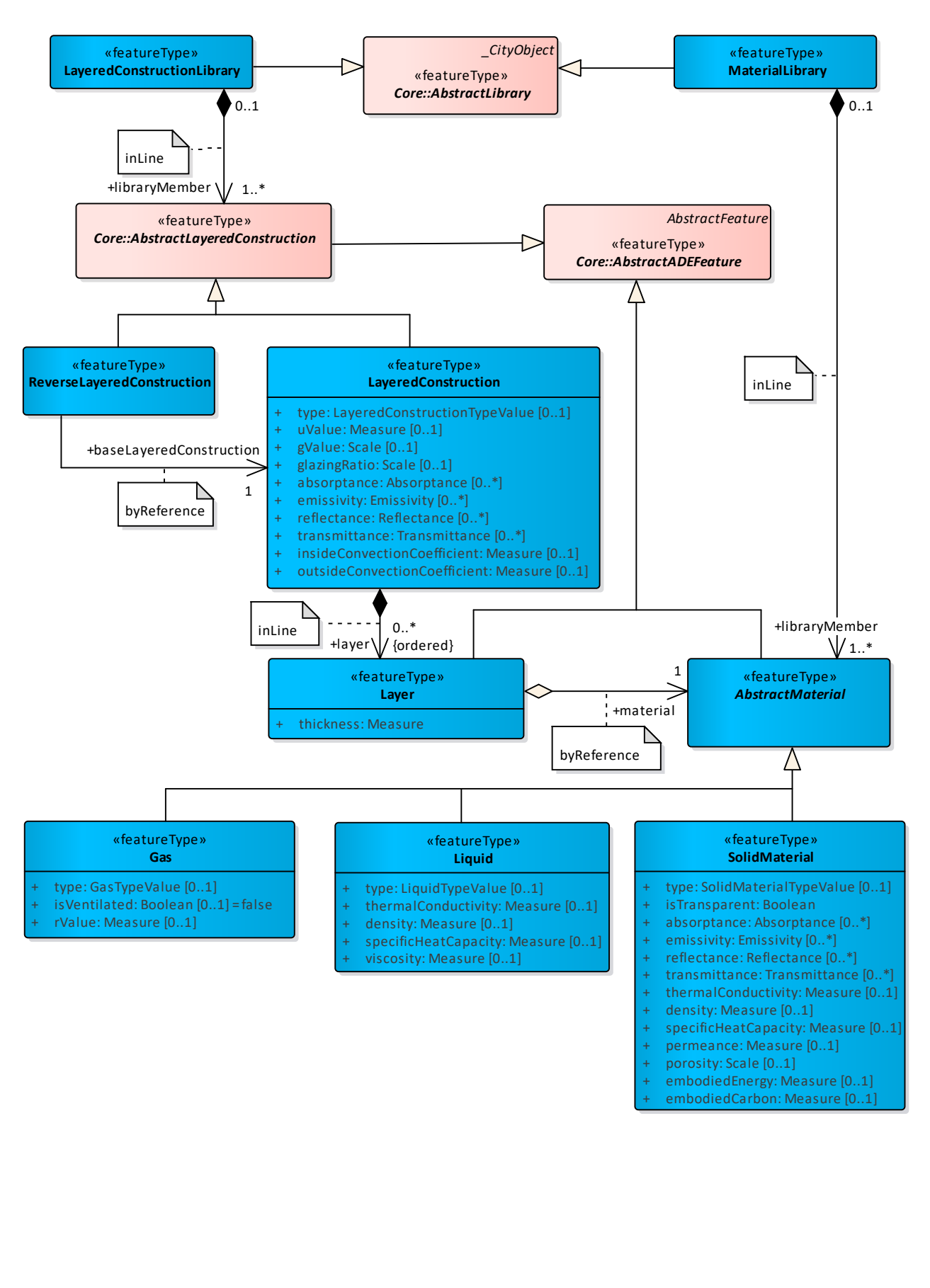


## Resources module

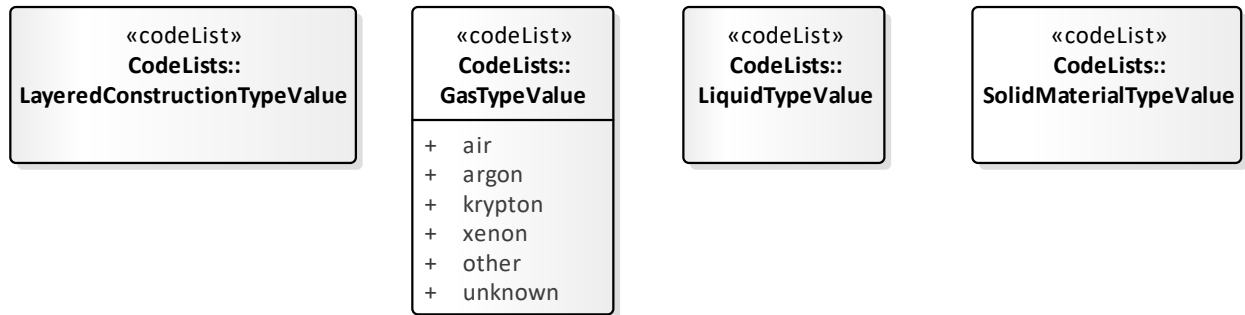


## Resources module: Codelists and Enumerations

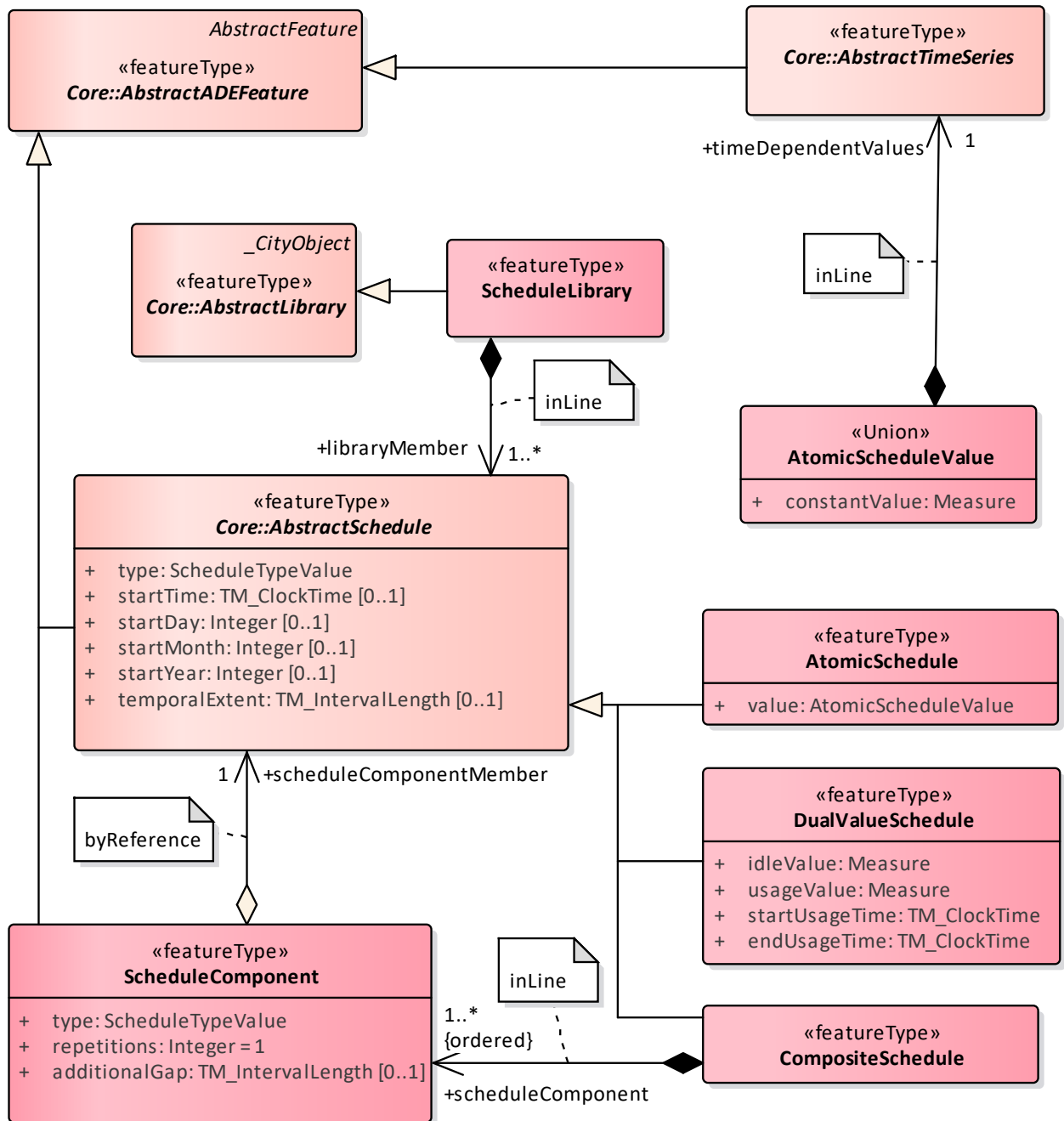




## Layered construction module: Codelists and Enumerations



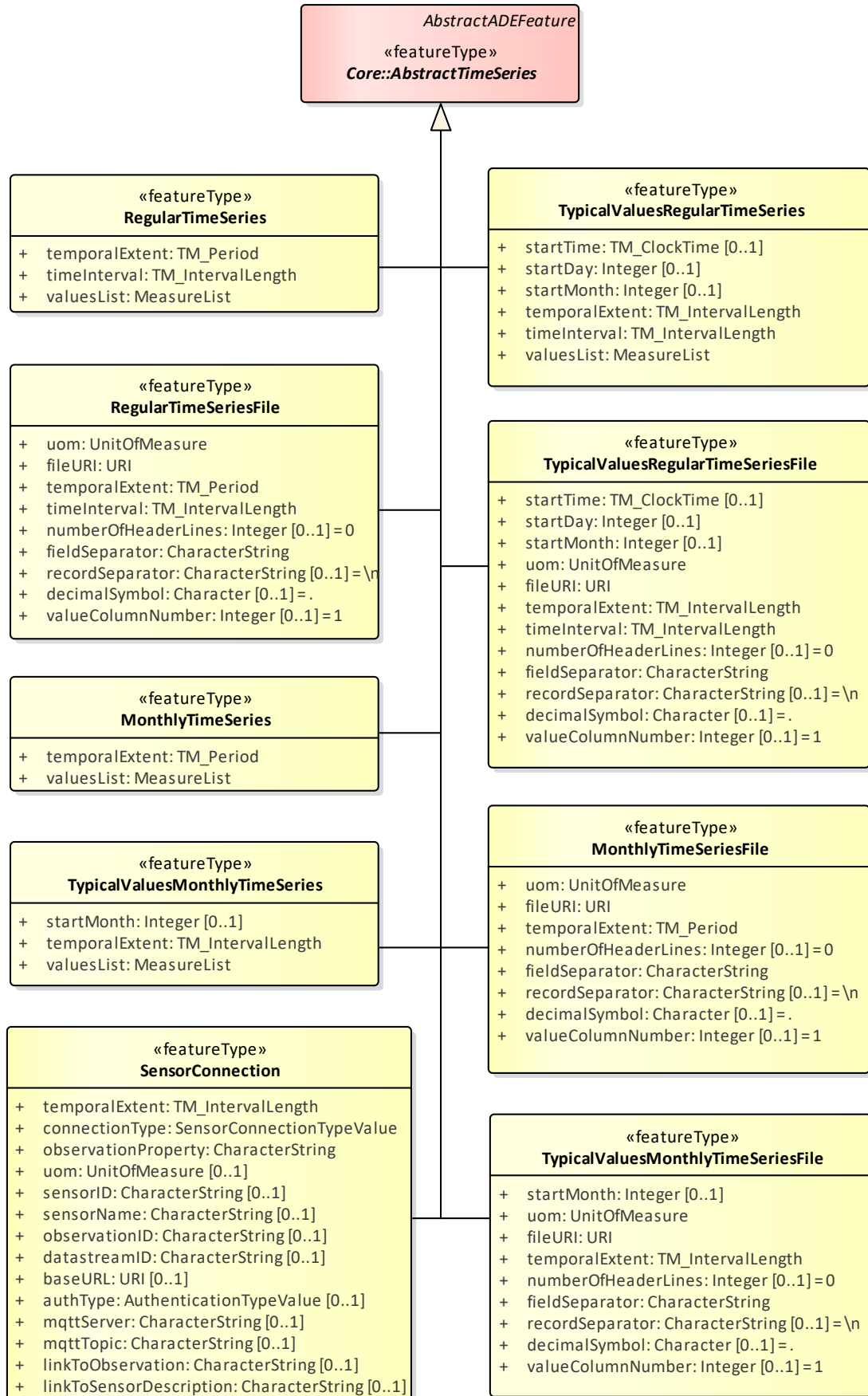
## Schedules module



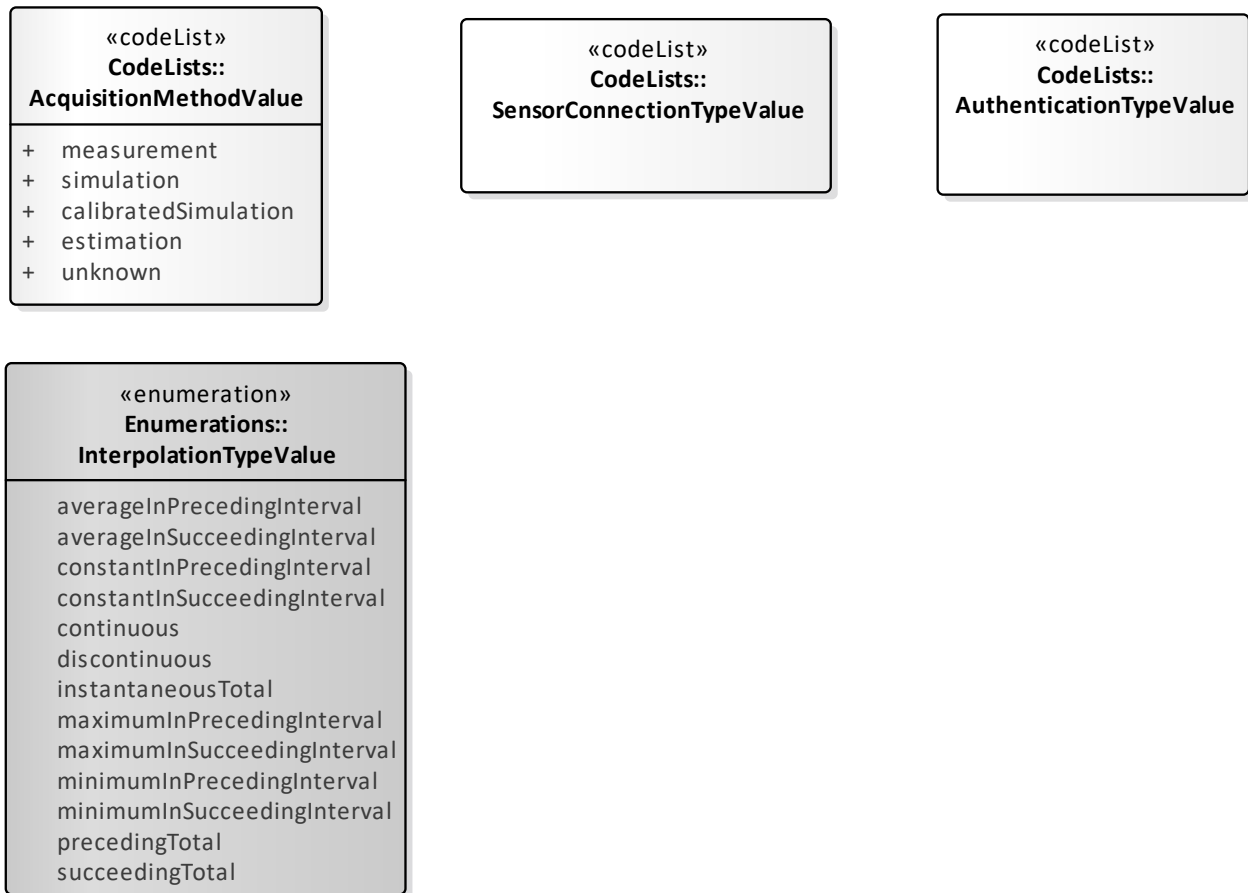
## Schedules module: Codelists and Enumerations

| «codeList»<br>CodeLists::<br>ScheduleTypeValue |               |
|------------------------------------------------|---------------|
| +                                              | hour          |
| +                                              | day           |
| +                                              | week          |
| +                                              | month         |
| +                                              | season        |
| +                                              | year          |
| +                                              | typicalDay    |
| +                                              | typicalWeek   |
| +                                              | typicalMonth  |
| +                                              | typicalSeason |
| +                                              | typicalYear   |
| +                                              | monday        |
| +                                              | tuesday       |
| +                                              | wednesday     |
| +                                              | thursday      |
| +                                              | friday        |
| +                                              | saturday      |
| +                                              | sunday        |
| +                                              | weekDay       |
| +                                              | weekendDay    |
| +                                              | beerDay       |
| +                                              | holiday       |
| +                                              | january       |
| +                                              | february      |
| +                                              | march         |
| +                                              | april         |
| +                                              | may           |
| +                                              | june          |
| +                                              | july          |
| +                                              | august        |
| +                                              | september     |
| +                                              | october       |
| +                                              | november      |
| +                                              | december      |
| +                                              | spring        |
| +                                              | summer        |
| +                                              | autumn        |
| +                                              | winter        |
| +                                              | other         |

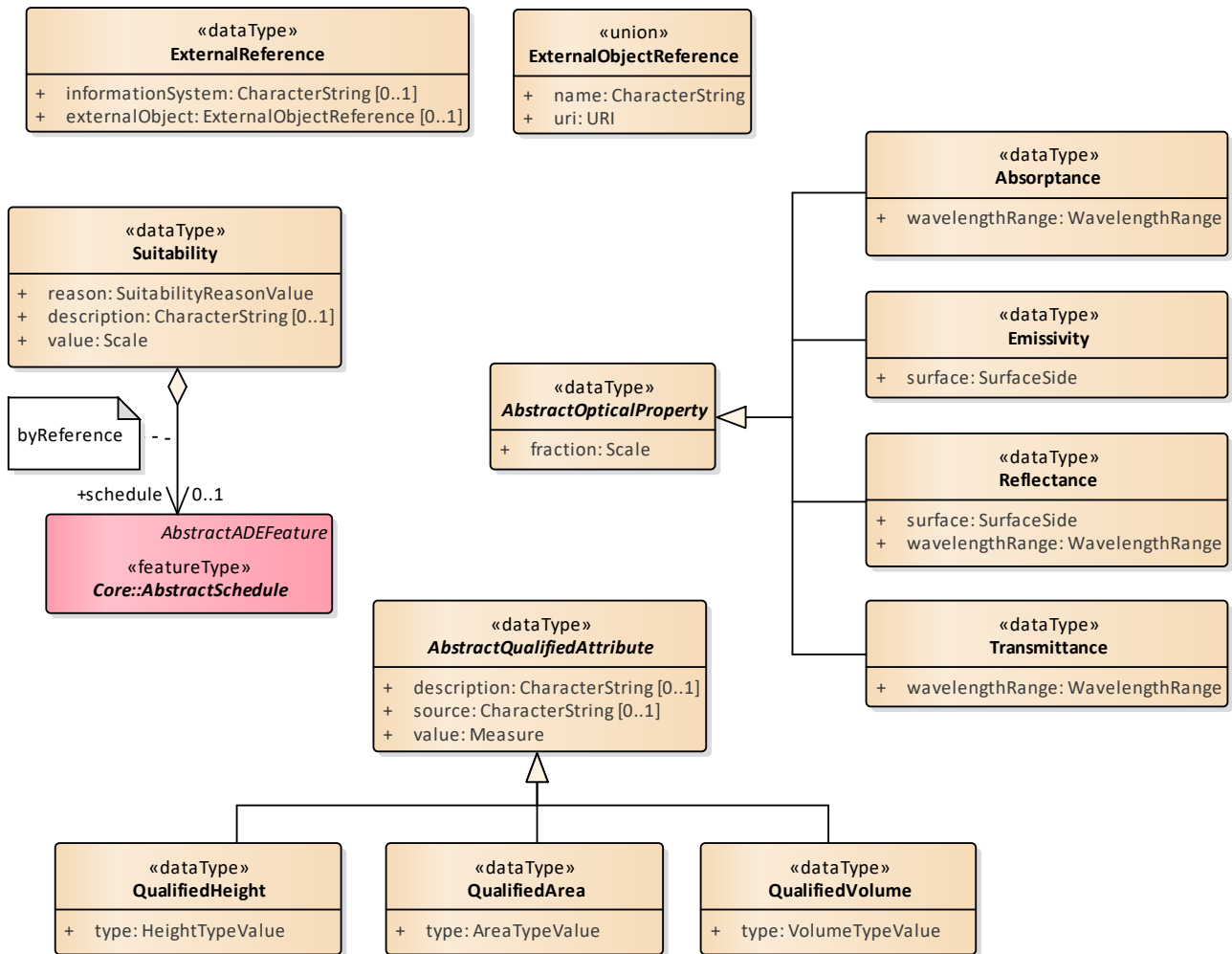
## Time series module



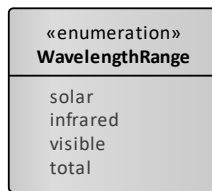
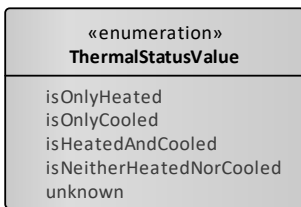
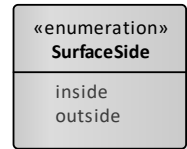
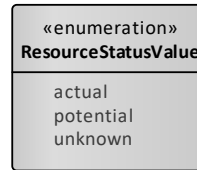
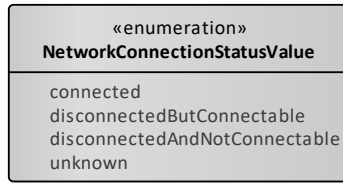
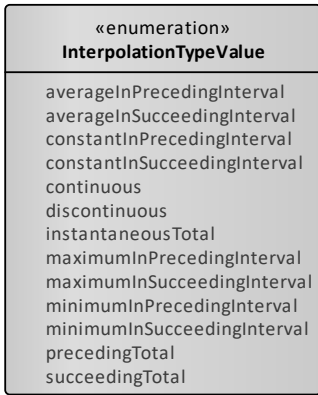
## Time series module: Codelists and Enumerations



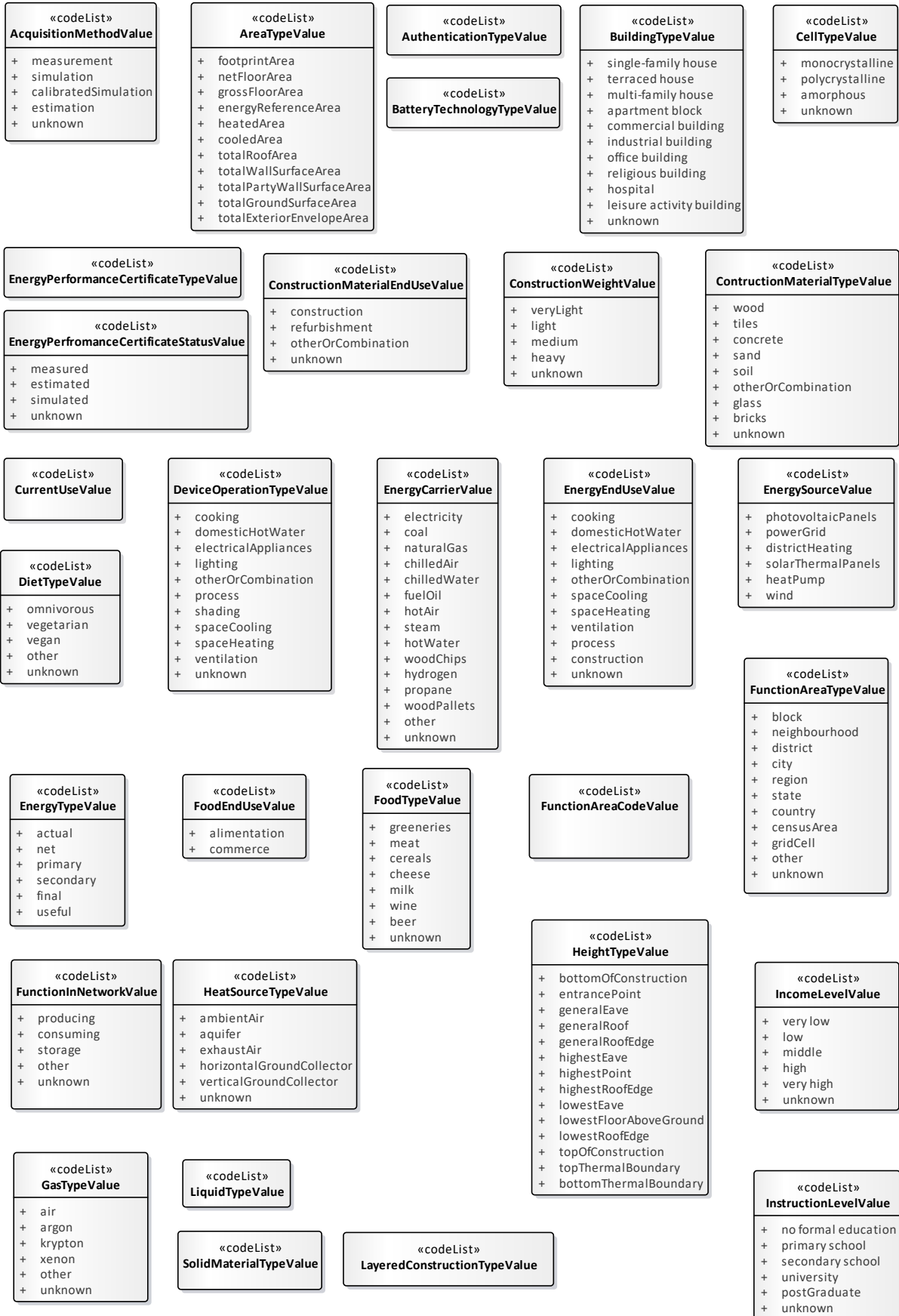
## Supporting classes: DataTypes

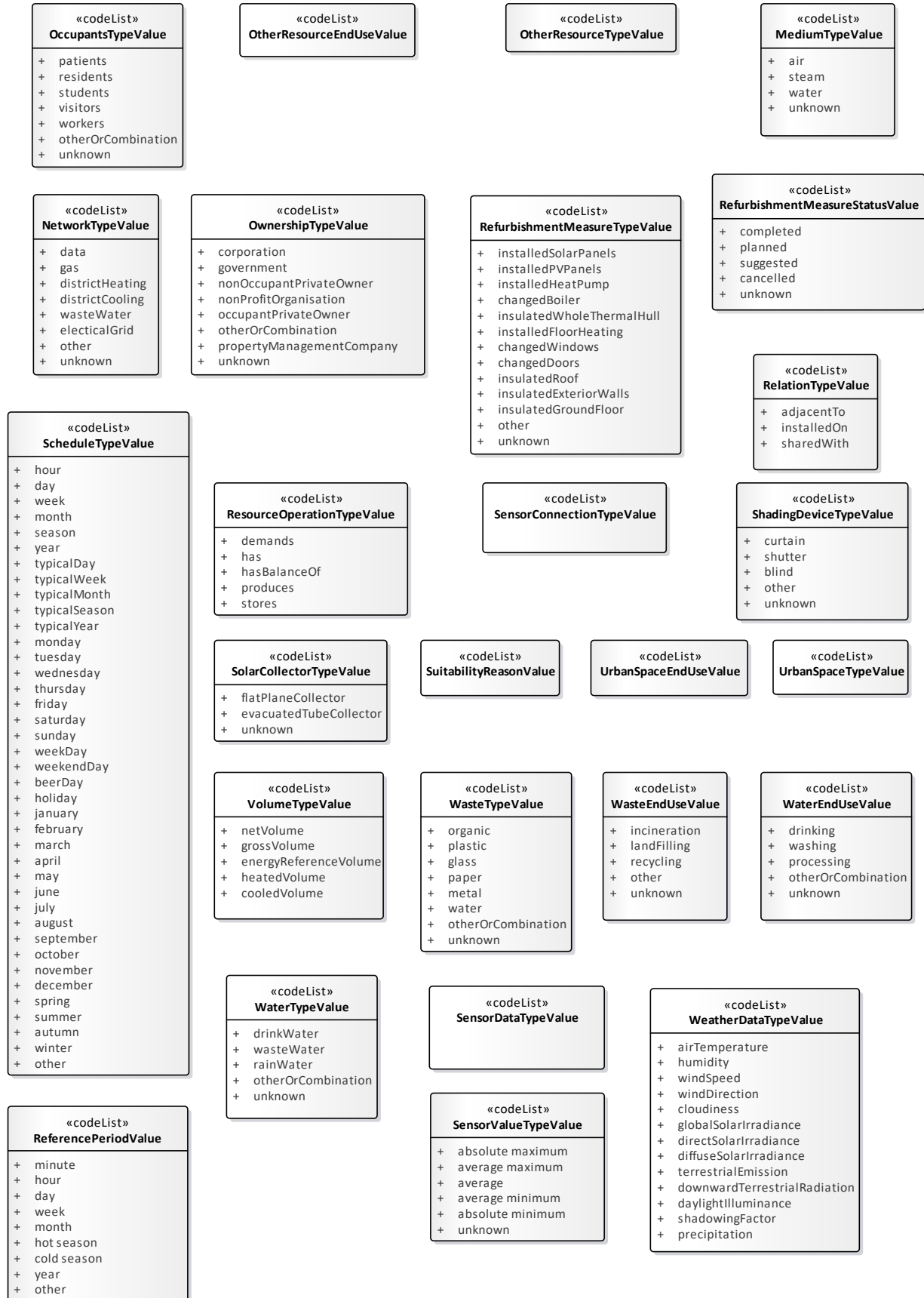


## All Enumerations



## All Codelists





## Changelog

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### Version 3.0 beta 8

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**2026-02-02**

**Data Type package:**

- Added class **ExternalReference**
- Added class **ExternalObjectReference**

**Building package:**

- Class **ADE \_AbstractBuilding**
  - Added property *bldNumberOfBuildingUnits*
- Class **\_AbstractZone**
  - Added property *bldNumberOfBuildingUnits*

**2025-11-25**

**CodeLists package:**

- Added class **WeatherDataTypeValue**
- Added class **GasTypeValue**
- Added class **LiquidTypeValue**
- Added class **SolidMaterialTypeValue**
- Added class **LayeredConstructionTypeValue**
- Added class **RefurbishmentMeasureStatusValue**
- Added class **EnergyPerformanceCertificateStatusValue**

**Data Type package:**

- Added class **Absorptance**

**Building package:**

- Class **RefurbishmentMeasure**
  - Added property *status*
- Class **EnergyPerfromanceCertificate**
  - Added property *status*
- Class **ADE \_AbstractBuilding**
  - Added property *bldOwnerName*
  - Added property *bldOwnershipType*

**Time series package:**

- Added class **MonthlyTimeSeries**
- Added class **MonthlyTimeSeriesFile**
- Added class **TypicalValuesMonthlyTimeSeries**
- Added class **TypicalValuesMonthlyTimeSeriesTile**
- Renamed class **TypicalValuesTimeSeries** to **TypicalValuesRegularTimeSeries**
- Renamed class **TypicalValuesTimeSeriesFile** to **TypicalValuesRegularTimeSeriesFile**

**Layered Construction package:**

- Class **LayeredConstructionMaterial**
  - Added property *type*
  - Added property *absorptance*
- Class **SolidMaterial**
  - Added property *type*
  - Added property *isTransparent*
  - Added property *absorptance*

- Added property *emissivity*
- Added property *reflectance*
- Added property *transmittance*
- Class **Liquid**
  - Added property *type*
- Class **Gas**
  - Added property *type*

#### WeatherStation package:

- Added class **WeatherData**

#### 2025-11-21

#### Core package:

- Class **AbstractADEFeature**
  - Added property *identifier*
  - Added property *validFrom*
  - Added property *validTo*
- Class **ADE\_CityObject**
  - Added property *permanentIdentifier*
  - Added property *validFrom*
  - Added property *validTo*
- Added class **AbstractFeatureWithLifeSpan**
- Class **UtilityNetworkConnection**
  - Class is now derived from class **AbstractFeatureWithLifeSpan**
- Class **AbstractResource**
  - Class is now derived from class **AbstractFeatureWithLifeSpan**

#### Building package:

- Class **EnergyPerformanceCertificate**
  - Class is now derived from class **AbstractFeatureWithLifeSpan**
  - Dropped property *issueDate*
  - Dropped property *expirationDate*
- Class **RefurbishmentMeasure**
  - Class is now derived from class **AbstractFeatureWithLifeSpan**
  - Renamed property *startDate* to *refurbishmentStartDate*
  - Renamed property *endDate* to *refurbishmentEndDate*

#### Schedules package:

- Class **ScheduleComponent**
  - Class is now derived from class **AbstractADEFeature**

#### Layered Construction package:

- Class **AbstractMaterial**
  - Class is now derived from class **AbstractADEFeature**
- Class **Layer**
  - Class is now derived from class **AbstractADEFeature**

#### 2025-11-19

#### CodeLists package:

- Renamed class **WeatherDataTypeValue** to **SensorDataTypeValue**
- Renamed class **WeatherValueTypeValue** to **SensorValueTypeValue**
- Added class **ReferencePeriod**

#### Core package:

- Renamed class **AbstractBuildingPartition** to **AbstractCityObjectSpace**
- Added class **AbstractADEFeature**
- Renamed class **WeatherData** to **SensorData**

- Class is now derived from class **AbstractADEFeature**
- Class **AbstractLayeredConstruction**
  - Class is now derived from class **AbstractADEFeature**
- Class **AbstractSchedule**
  - Class is now derived from class **AbstractADEFeature**
  - Dropped property *libraryCode*
- Class **AbstractTimeSeries**
  - Class is now derived from class **AbstractADEFeature**
- Class **AbstractResource**
  - Class is now derived from class **AbstractADEFeature**
  - Added property *timeDependentCosts*
  - Added property *timeDependentYields*
  - Added property *referencePeriod*
- Class **DeviceOperation**
  - Class is now derived from class **AbstractADEFeature**
- Class **EnergyPerformanceCertificate**
  - Class is now derived from class **AbstractADEFeature**
  - Class moved to Building module
- Class **RefurbishmentMeasure**
  - Class is now derived from class **AbstractADEFeature**
  - Dropped property *libraryCode*
  - Class moved to Building module
- Class **UtilityNetworkConnection**
  - Class is now derived from class **AbstractADEFeature**
- Class **ADE\_CityObject**
  - Renamed relation *layeredConstruction* to *libraryLayeredConstruction*
  - Added relation *layeredConstruction*
- Class **ADE\_AbstractBuilding**
  - Class moved to Building module
- Dropped class **AbstractBuildingZone**
- Dropped class **AbstractUsageZone**
- Dropped class **AbstractThermalZone**

#### Building physics package:

Dropped. Some classes reworked and moved to new module Building

#### Occupancy package:

Dropped. Some classes reworked and moved to new module Building

#### Building package:

This module is new. It reworks and merges the previous Building physics module and Occupancy module. Additionally, it incorporates also class *ADE\_AbstractBuilding* and its subclasses.

- Class **ADE\_AbstractBuilding**
  - Class is moved from Core module
  - Dropped property *usageZone*
  - Dropped property *thermalZone*
  - Added property *occupiedBy*
  - Added property *zone*
- Class **EnergyPerformanceCertificate**
  - Class is moved from Core module
  - Class is derived from class **AbstractADEFeature**
- Class **RefurbishmentMeasure**
  - Class is moved from Core module
  - Class is derived from class **AbstractADEFeature**
  - Dropped property *libraryCode*
- Class **Occupants**
  - Class is taken over from former Occupancy module
  - Class is now derived from class **AbstractADEFeature**

- Renamed relation *occupancyRate* to *occupancySchedule*
- Added class **AbstractBuildingSpace**
- Class **BuildingUnit**
  - Class is taken over from former Occupancy module
  - Class is now derived from class **AbstractBuildingSpace**
- Added class **AbstractZone**
- Added class **Zone**
- Added class **ZonePart**
- Class **ADE \_BoundarySurface**
  - Class is taken over from former Building physics module
  - Added property *bdgBdrySurflsShared*
  - Dropped property *bdgBdrySurflsAdiabatic*
- Added class **AbstractZoneBoundarySurface**
- Added class **ZoneAtticFloorSurface**
- Added class **ZoneBasementCeilingSurface**
- Added class **ZoneExteriorWallSurface**
- Added class **ZoneGroundSurfaceSurface**
- Added class **ZoneIntermediateFloorSurface**
- Added class **ZoneRoofSurface**
- Added class **ZoneUndergrounSurface**
- Added class **ZoneClosureSurface**
- Added class **ADE \_Opening**
  - Class is taken over from former Building physics module
- Added class **ADE AbstractZoneOpening**
- Added class **ZoneDoor**
- Added class **ZoneWindow**

#### Device package:

- Added class **GenericStorageDevice**

#### Layered Construction package:

- Class **MaterialLibrary**
  - Changed multiplicity of property *libraryMember* to 1..\*
- Class **LayeredConstructionLibrary**
  - Changed multiplicity of property *libraryMember* to 1..\*
- Class **LayeredConstruction**
  - Dropped property *libraryCode*
  - Added property *insideConvectionCoefficient*
  - Added property *outsideConvectionCoefficient*
- Class **AbstractMaterial**
  - Dropped property *libraryCode*
- Added class **Liquid**

#### Schedules package:

- Class **AtomicSchedule**
  - Dropped property *timeSeries*
  - Added property *value*
- Added class **AtomicScheduleValue**
- Class **ScheduleLibrary**
  - Changed multiplicity of property *libraryMember* to 1..\*

#### Urban function area package:

- Class **UrbanFunctionArea**
  - Add property *area*

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#### Version 3.0 beta 7

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**2025-10-14**

No changes, except for the version number from 2.0 to 3.0

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**Version 2.0 beta 7**

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**2025-06-25****Schedules package:**

- Class **ScheduleLibrary**
  - Changed relation type of *libraryMember* to composition

**Layered Construction package:**

- Class **LayeredConstructionLibrary**
  - Changed relation type of *libraryMember* to composition
- Class **LayeredConstruction**
  - Added property *gValue*
- Class **MaterialLibrary**
  - Changed relation type of *libraryMember* to composition

**2025-05-30****Schedules package:**

- Class **AtomicSchedule**
  - Changed multiplicity of relation *timeSeries* to [0..1]
  - Added property *constantValue*

**2025-05-13****Building physics package:**

- Class **ThermalZone**
  - Changed relation type of *thermalBoundary* to composition

**2025-05-07****Core package:**

- Class **ADE\_CityObject**
  - Changed relation type of *device* to aggregation
  - Changed relation type of *weatherData* to composition
- Class **AbstractThermalZone**
  - Changed relation type of *usageZone* to association
- Class **AbstractUsageZone**
  - Changed relation type of *buildingUnit* to association
- Class **AbstractResource**
  - Made relation to **AbstractTimeSeries** explicit and set to composition
- Class **DeviceOperation**
  - Made relation to **AbstractSchedule** explicit and set to aggregation
- Class **WeatherData**
  - Made relation to **AbstractTimeSeries** explicit and set to composition

**Occupancy package:**

- Class **UsageZone**
  - Changed relation type of *occupiedBy* to aggregation
  - Made all relations to **AbstractSchedule** explicit and set to aggregation
- Class **BuildingUnit**
  - Changed relation type of *occupiedBy* to aggregation

- Class **Occupants**
  - Made relation to **AbstractSchedule** explicit and set to aggregation

#### Layered Construction package:

- Class **LayeredConstruction**
  - Changed relation type of *layer* to composition
- Class **Layer**
  - Changed relation type of *material* to aggregation

#### Schedules package:

- Class **ScheduleComponent**
  - Changed relation type of *scheduleComponentMember* to aggregation

#### 2025-05-01

#### Core package:

- Class **ADE \_AbstractSchedule**
  - Moved from package Schedules to Core
- Class **ADE \_AbstractTimeSeries**
  - Moved from package Time Series to Core

#### Occupancy package:

- Class **Occupants**:
  - Added property *averageDietType*
  - Added property *averageIncomeLevel*
  - Added property *averageInstructionLevel*

#### CodeLists package:

- Added class **DietTypeValue**
- Added class **InstructionLevelValue**
- Added class **IncomeLevelValue**

#### 2025-04-23

#### CodeLists package:

- Added values to class **AreaTypeValue**
- Renamed class **HeightReferenceValue** to **HeightTypeValue**
- Added class **SuitabilityReasonValue**
- Added class **UrbanSpaceTypeValue**
- Added class **UrbanSpaceEndUseValue**

#### DataTypes package:

- Added class **Suitability**

#### Core package:

- Class **ADE \_CityObject**
  - Added property *suitability*
- Class **ADE \_AbstractBuilding**
  - Renamed property *bdgFloorArea* to *bdgArea*
  - Renamed property *bdgHeightAboveGround* to *bdgHeight*
- Class **AbstractBuildingPartition**
  - Renamed property *floorArea* to *area*

#### Resources package:

- Added class **UrbanSpace**

#### 2025-04-12

#### Core package:

- Added class **AbstractLibrary**

#### Layered Construction package:

- Added class **LayeredConstructionLibrary**
- Added class **MaterialLibrary**

#### Schedules package:

- Added class **ScheduleLibrary**

#### CodeLists package:

- Class **FunctionInNetwork** renamed to **FunctionInNetworkValue**

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### **Version 2.0 beta 6**

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**2025-02-06**

#### DataTypes package:

- Dropped class **HeatExchangeType**
- Dropped class **OpticalProperties**

#### Core package:

- Class **ADE\_CityObject**:
  - Renamed property from *operationSchedule* to *deviceOperation*
- Class **AbstractDevice**:
  - Changed type of property *heatDissipation* to gml:Measure
  - Added property *heatDissipationConvectiveFraction*
  - Added property *heatDissipationLatentFraction*
  - Added property *heatDissipationRadiantFraction*

#### Occupancy package:

- Class **UsageZone**:
  - Renamed property *averageInternalGains* to *internalGains* and changed type to gml:Measure
  - Added property *internalGainsConvectiveFraction*
  - Added property *internalGainsLatentFraction*
  - Added property *internalGainsRadiantFraction*
- Class **Occupants**:
  - Changed type of property *heatDissipation* to gml:Measure
  - Added property *heatDissipationConvectiveFraction*
  - Added property *heatDissipationLatentFraction*
  - Added property *heatDissipationRadiantFraction*

#### Layered Construction package:

- Class **LayeredConstruction**:
  - Dropped property *opticalProperties*
  - Added property *glazingRatio*
  - Added property *emissivity*
  - Added property *reflectance*
  - Added property *transmittance*
- Class **Layer**:
  - Changed multiplicity of property *thickness* to [1]
- Class **AbstractMaterial**:
  - Added property *libraryCode*

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### **Version 2.0 beta 5**

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#### Resources package:

No changes since beta 4

#### Weather Station package:

No changes since beta 4

#### 2025-01-30

##### Packages

- Created TimeSeries package and moved all related classes into it
- Created Schedule package and moved all related classes into it
- Dropped Supporting classes package

##### CodeLists package:

- Class **CertificateTypeValue** renamed to **EnergyPerformanceCertificateTypeValue**

#### 2025-01-28

##### CodeLists package:

- Added class **ScheduleTypeValue**

##### Enumerations package:

- Dropped class **DayTypeValue**

##### Supporting classes package:

- Dropped class **TimeSeriesSchedule**
- Dropped class **DailyPatternSchedule**
- Dropped class **PeriodOfYear**
- Dropped class **DailySchedule**
- Added class **AtomicSchedule**
- Added class **CompositeSchedule**
- Added class **ScheduleComponent**
- Class **AbstractSchedule**:
  - Added properties *type*, *startTime*, *startDay*, *startMonth*, *startYear*, *temporalExtent*

#### 2025-01-22

##### Devices package:

- Class **AbstractSolarCollector**:
  - Added properties *azimuth* and *inclination*
  - Dropped property *surfaceGeometry*
  - Added properties *lod2MultiSurface* and *lod3MultiSurface*

#### 2025-01-13

##### CodeLists package:

- Dropped class **ThermalBoundaryTypeValue**
- Added class **ShadingDeviceTypeValue**
- Added class **WeatherValueTypeValue**

##### DataTypes package:

- Moved class **PeriodOfYear** from package DataTypes to Supporting classes
- Moved class **DailySchedule** from package DataTypes to Supporting classes
- Class **AbstractQualifiedAttribute**:
  - Added properties *source* and *value*
- Classes **QualifiedHeight**, **QualifiedArea**, **QualifiedVolume**:
  - Dropped property *value*

**Supporting classes package:**

- Class **AbstractTimeSeries**:
  - Moved back from package Core to Supporting classes
  - Changed multiplicity of property *acquisitionMethod* to [1]
  - Changed multiplicity of property *interpolationType* to [1]
- Class **TypicalValuesTimeSeries**:
  - Added properties *startTime*, *startDay*, *startMonth*
- Class **TypicalValuesTimeSeriesFile**:
  - Added properties *startTime*, *startDay*, *startMonth*
- Added class **DualValueSchedule**:
- Class **PeriodOfYear**:
  - Changed stereotype from *dataType* to *featureType*
  - Added properties *startDay*, *startMonth*, *endDay*, *endMonth*
  - Dropped property *period*
- Class **DailySchedule**:
  - Changed stereotype from *dataType* to *featureType*
  - Added property *repetition*
  - Renamed property from *schedule* to *timeSeries*

**Layered Construction package:**

- Added class **ReverseLayeredConstruction**
- Dropped class **LayerComponent**
- Class **Layer**:
  - Removed property *layerComponent*
  - Added property *thickness*
  - Added property *material*
- Class **SolidMaterial**:
  - Renamed property from *specificHeat* to *specificHeatCapacity*
  - (Re)added and renamed property from *conductivity* to *thermalConductivity*
  - (Re)added property *embodiedEnergy*
  - (Re)added property *embodiedCarbon*
  - (Re)added property *porosity*

**Core package:**

- Class **AbstractDevice**:
  - Class is renamed from **AbstractEnergySystem** to **AbstractDevice**
  - Added property *heatDissipation*
  - Added property *operationSchedule*
  - Added property *efficiencyIndicator*
  - Added property *installedPower*
  - Added property *nominalEfficiency*
- Class **AbstractTimeSeries**:
  - Moved back from package Core to Supporting classes
- Class (ADE) **\_CityObject**:
  - Renamed property from *energySystem* to *device*
- Class **OperationSchedule**:
  - Moved from package Devices to Core
- Class **WeatherData**:
  - Renamed property from *timeseries* to *timeDependentValues*
  - Renamed property from *singleValue* to *yearlyValue*
  - Renamed property from *singleValueType* to *valueType*
  - Changed type of property *valueType* from *CharacterString* to *CodeList*
- Added class **AbstractBuildingPartition**
- Classes **AbstractBuildingUnit**, **AbstractUsageZone**, **AbstractThermalZone**
  - Changed parent class

**Building physics package:**

- Dropped class **ThermalBoundary**

- Dropped class **ThermalOpening**
- Class **ThermalZone**
  - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
  - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
  - Removed property *volumeGeometry* (now inherited via **AbstractBuildingPartition**)
  - Removed property *boundedBy*
  - Added property *heatCapacity*
  - Added property *coincidesWithLod2Hull*
  - Added property *coincidesWithLod3Hull*
  - Added property *thermalBoundary*
- Added class **PartyWallSurface**
- Added class **BasementCeilingSurface**
- Added class **IntermediateFloorSurface**
- Added class **AtticFloorSurface**
- Class (ADE) **\_BoundarySurface**:
  - Renamed property from *bdgBdrySrfWindowToWallRatio* to *bdgBdrySrfOpeningToSurfaceRatio*
  - Renamed property from *bdgBdrySrfArea* to *bdgBdryTotalSurfaceArea*
  - Added property *bdgBdryOpaqueSurfaceArea*
  - Added property *bdgBdrySrfThickness*
  - Added property *bdgBdrySrfHeatCapacity*

#### Occupancy package:

- Dropped class **AbstractFacilities**
- Dropped class **DHWFacilities**
- Dropped class **LightingFacilities**
- Dropped class **ElectricalAppliances**
- Dropped class **GenericFacilities**
- Class **BuildingUnit**:
  - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
  - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
  - Removed property *equippedWith*
- Class **UsageZone**:
  - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
  - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
  - Removed property *volumeGeometry* (now inherited via **AbstractBuildingPartition**)
  - Removed property *equippedWith*

#### Devices package:

The former EnergySystems package is renamed to Devices package

- Class **OperationSchedule**:
  - Moved from package Devices to Core
- Dropped class **AbstractEnergyConversionSystem**
  - Now all subclasses are derived directly from **AbstractDevice**
- Dropped class **AbstractStorageSystem**
  - Now all subclasses are derived directly from **AbstractDevice**
- Class **GenericDevice**:
  - Renamed from class **GenericEnergyConversionSystem**
- Class **ElectricalStorageDevice**:
  - Renamed from class **PowerStorageSystem**
- Class **ThermalStorageDevice**:
  - Renamed from class **ThermalStorageSystem**
- Class **AbstractSolarCollector**:
  - Renamed from class **AbstractSolarEnergySystem**
- Class **PhotovoltaicThermalCollector**:
  - Renamed from class **PhotovoltaicThermalSystem**
- Class **PhotovoltaicCollector**:
  - Renamed from class **PhotovoltaicSystem**
- Class **SolarThermalCollector**:
  - Renamed from class **SolarThermalSystem**

- Class **GenericSolarCollector**:
  - Renamed from class **GenericSolarEnergySystem**
- Added class **LightingDevice**
- Added class **GenericElectricalDevice**
- Added class **MovableShadingDevice**

#### Urban Function Areas package:

- Class **UrbanFunctionArea**:
  - Removed property *rank*

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### Version 2.0 beta 4

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#### 2024-09-25

- Codelist **ThermalBoundaryTypeValue**
  - Renamed *outerWall* to *exteriorWall*
  - Renamed *intermediaryFloor* to *intermediateFloor*

#### 2024-09-20

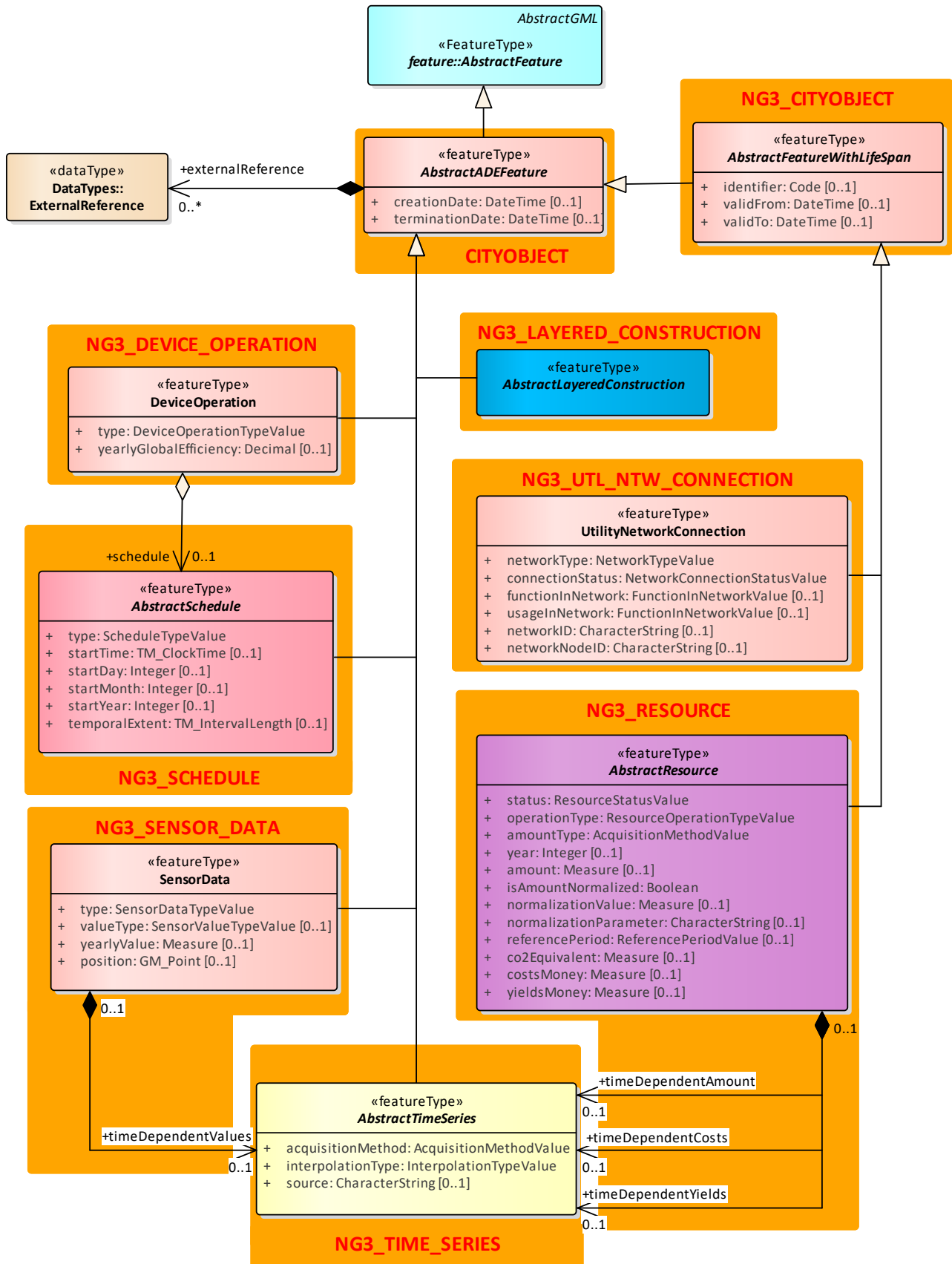
- Minor edits to the layout of the UML diagrams
- Polished Codelist **ThermalBoundaryTypeValue**

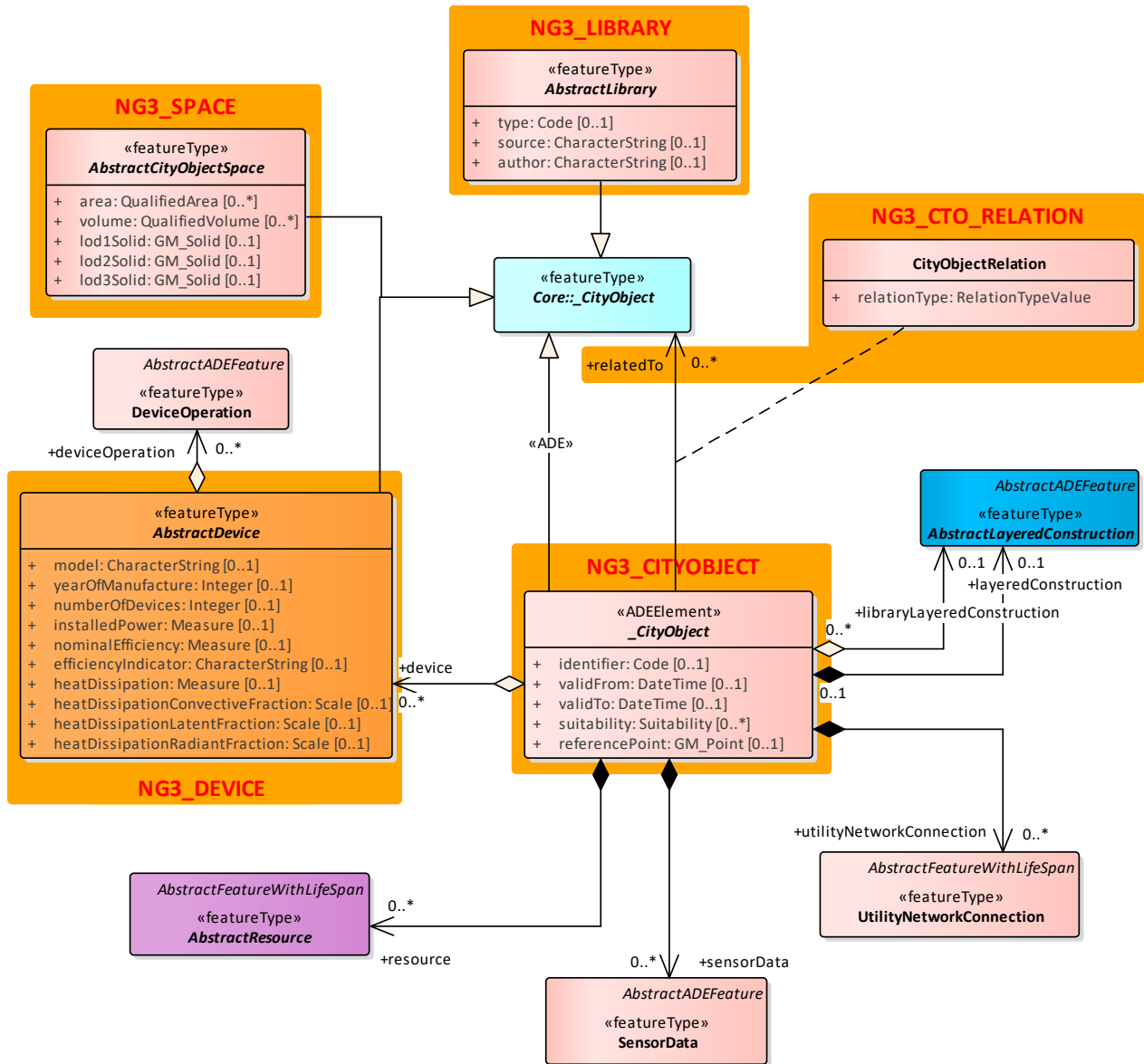
#### 2024-08-12

- Class **ADE CityObject**:
  - Renamed association *connectedTo* to *utilityNetworkConnection*
  - Remodelled association class **CityObjectRelation**
- Class **AbstractTimeSeries**:
  - Moved to Core module
- Class **WeatherData**:
  - renamed property *valuesList* to *timeSeries*
  - made the association to **AbstractTimeSeries** explicit
- Added some values to codelist **RelationTypeValue**
- Class **TimeSeriesSchedule**:
  - Changed association to **AbstractTimeSeries** to Composition
- Class **DailySchedule**:
  - Changed association to **AbstractTimeSeries** to Composition
- Class **SensorConnection**:
  - Changed type of *uom* to *UnitOfMeasure*
  - Renamed *linkTosensortDescription* to *linkToSensorDescription*
- Class **AbstractSolarEnergySystem**:
  - Removed association *installedOn* (now taken over by **CityObjectRelation**)
- Class **Energy**:
  - Changed multiplicity of *EnergyCarrier* to [0..1]
- Class **Waste**:
  - Changed multiplicity of *isDangerous* and *isRecyclable* to [0..1]

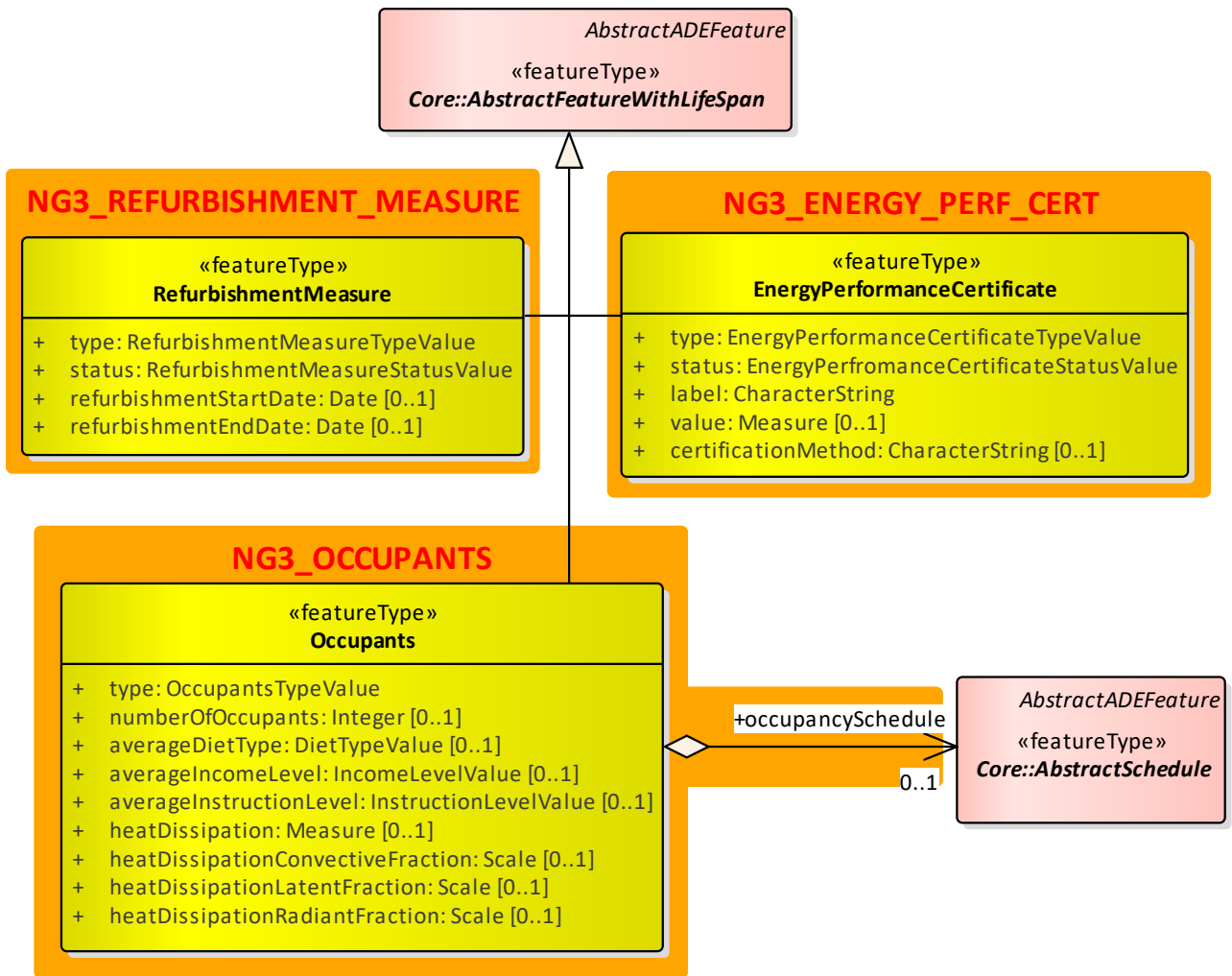
# **Part 3**

## **OO to ER mapping**

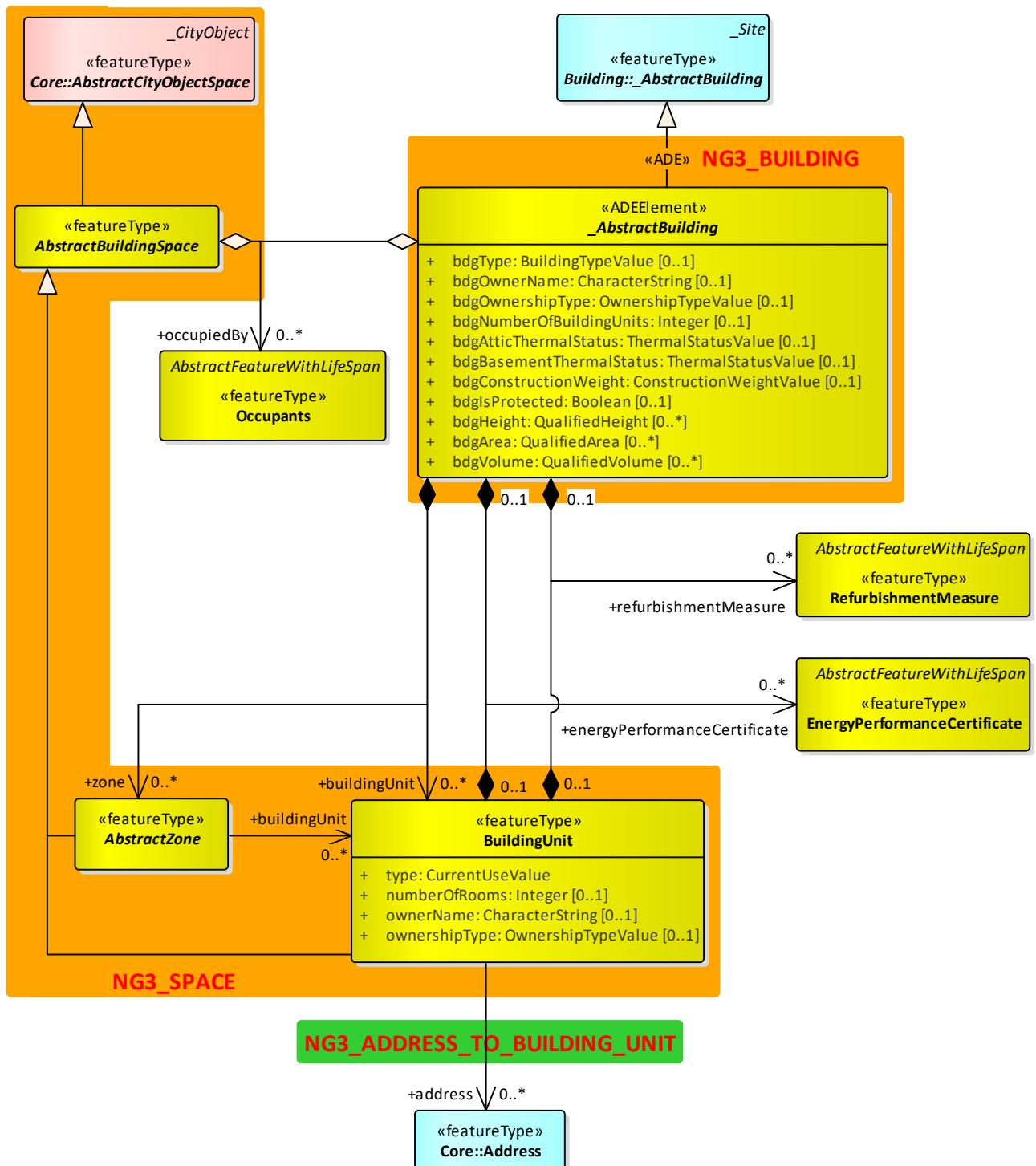
Core module: *AbstractADEFeature* and related classes

Core module: *ADE\_CityObject* and related classes

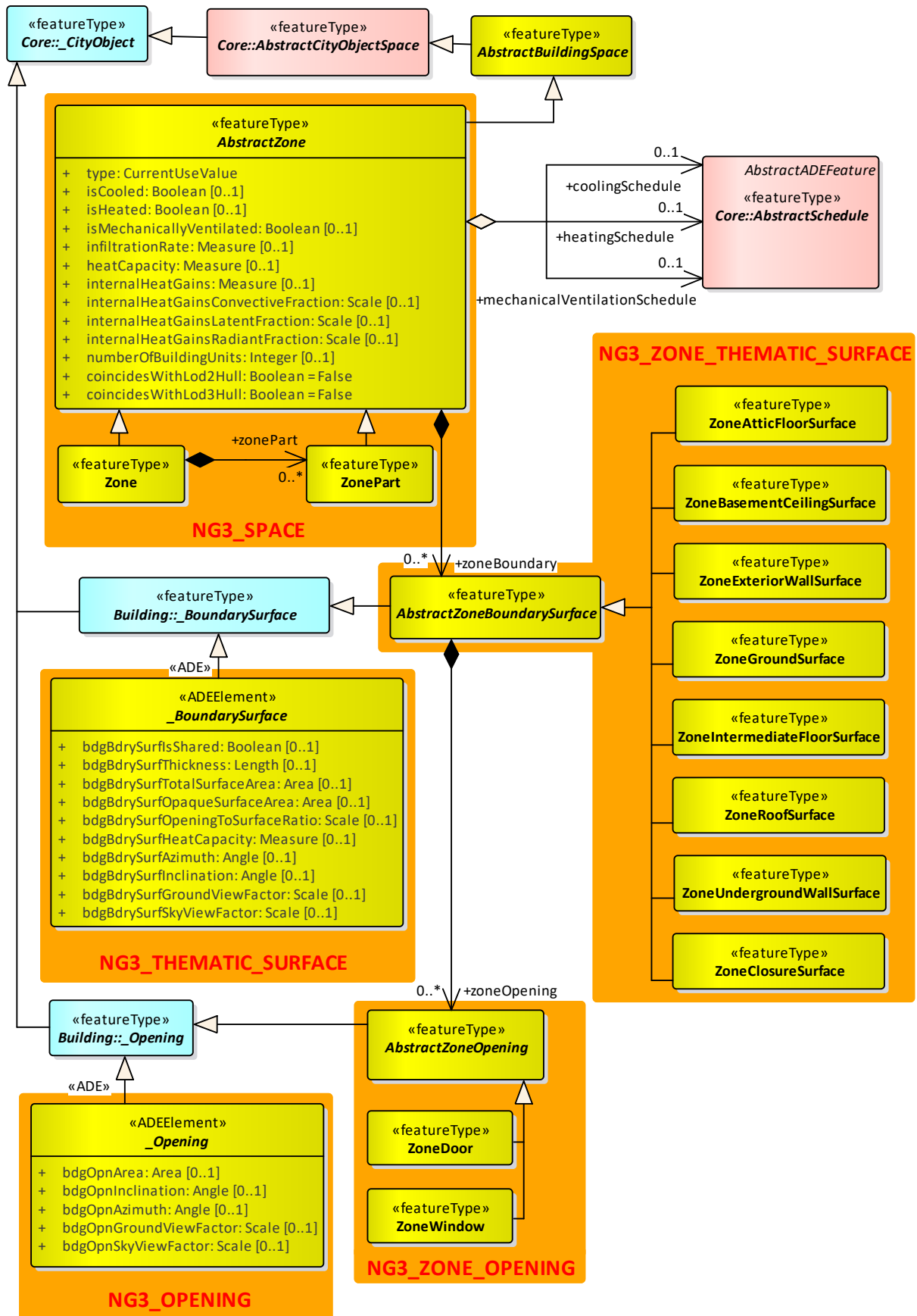
## Building module: Feature classes



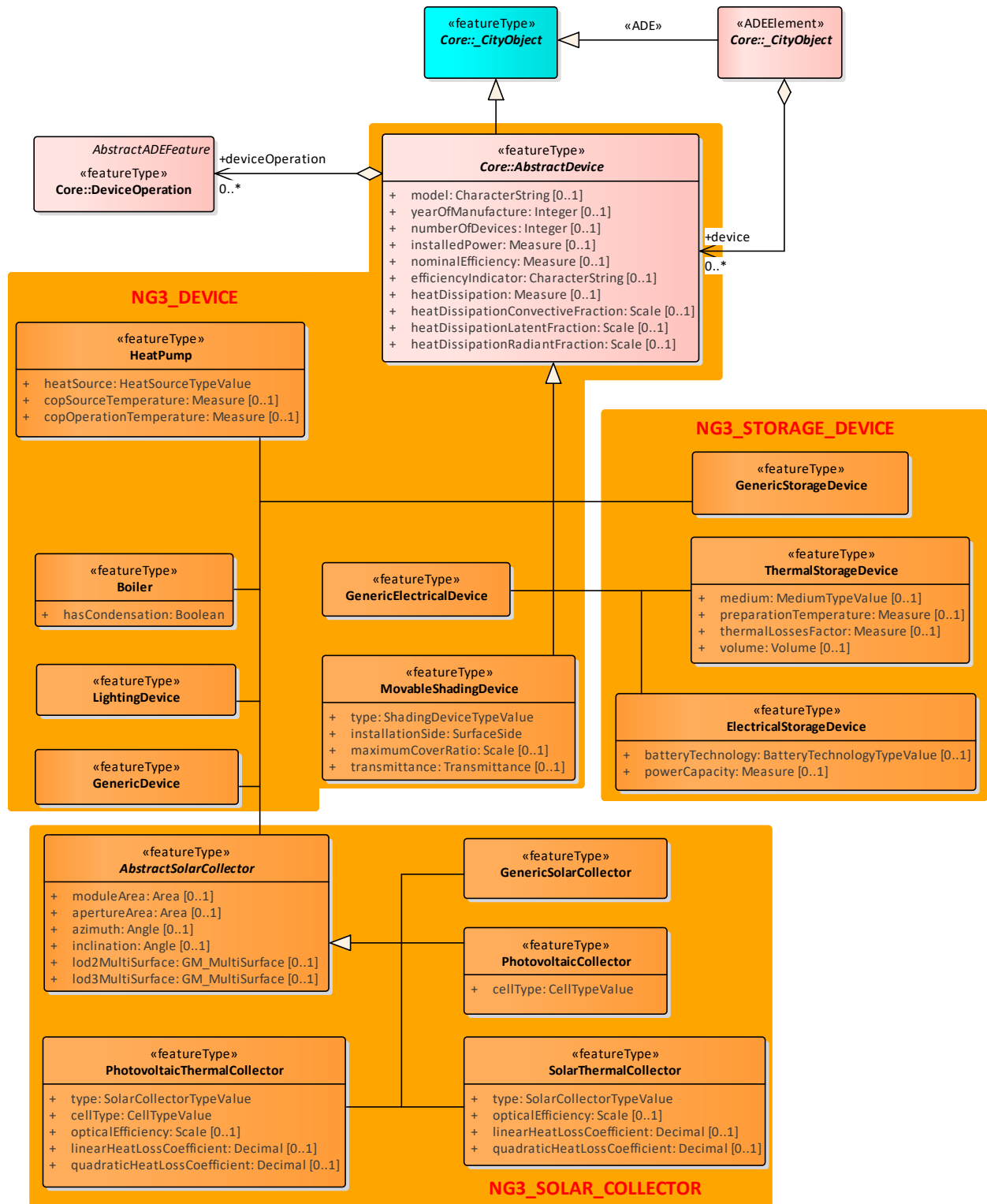
## Building module: ADE *\_AbstractBuilding* and related classes



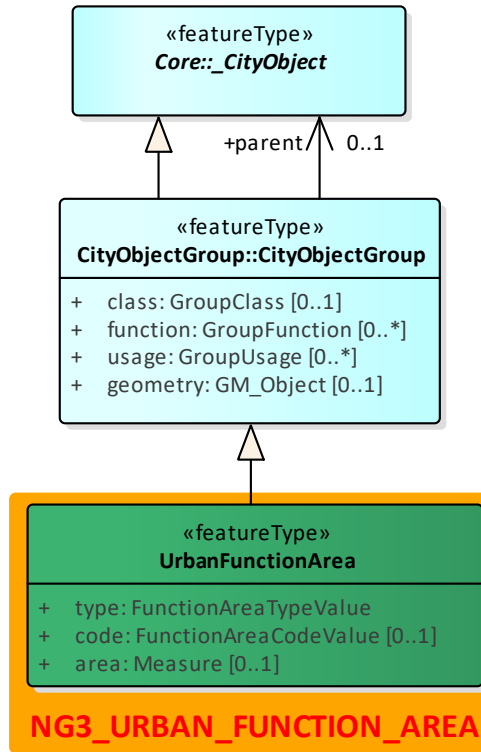
## Building module: *AbstractZone* and related classes



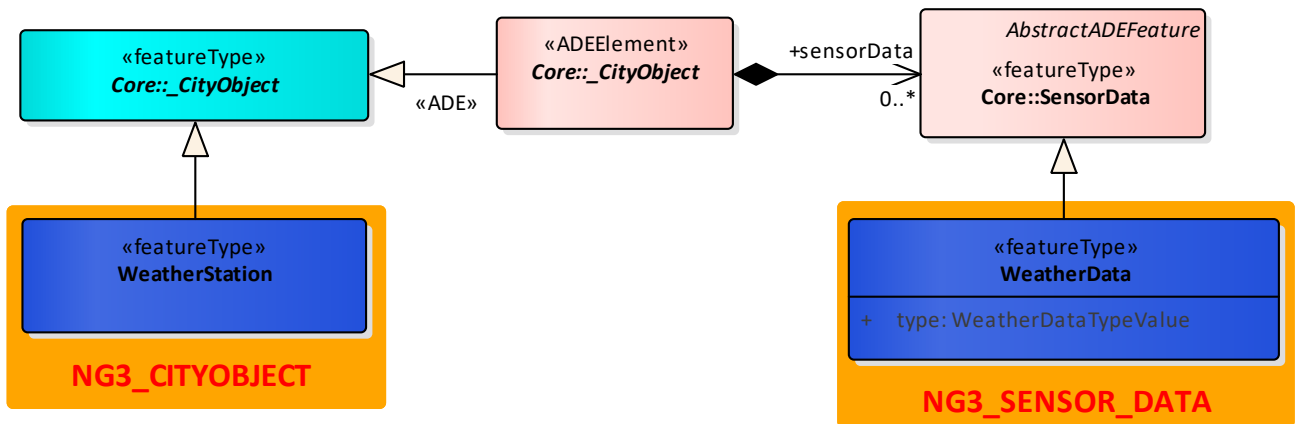
## Devices module



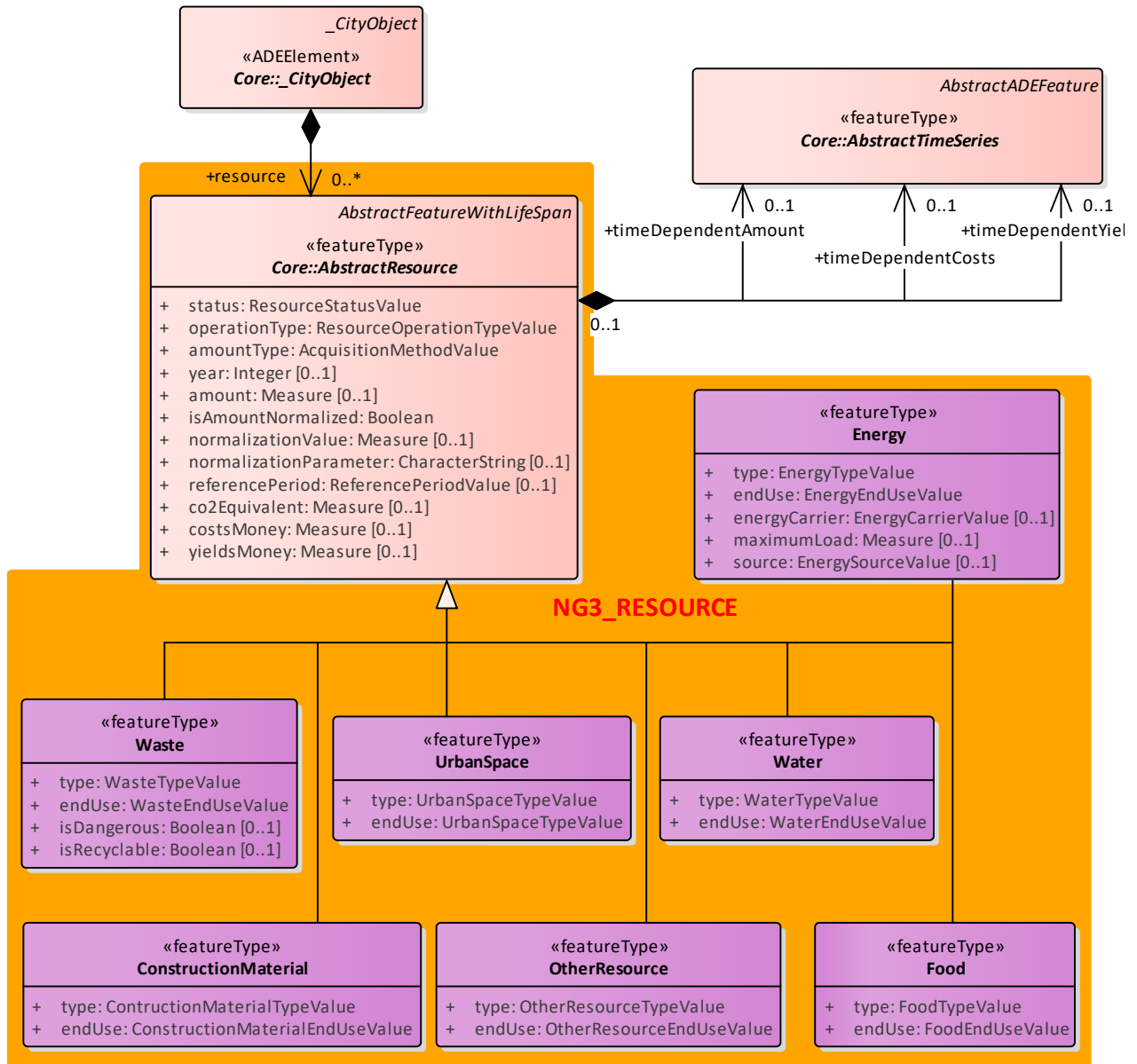
### Urban function areas module



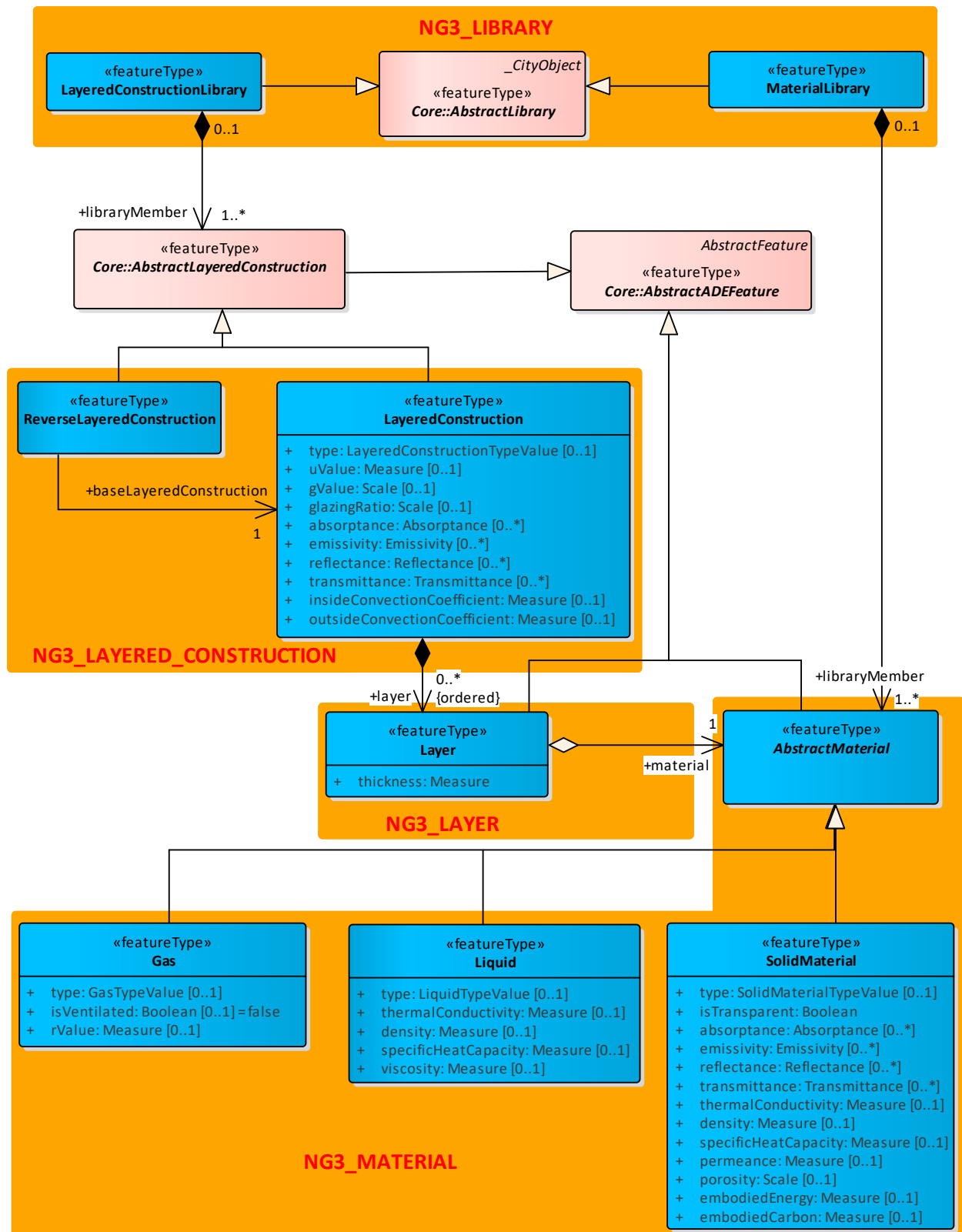
### Weather station module



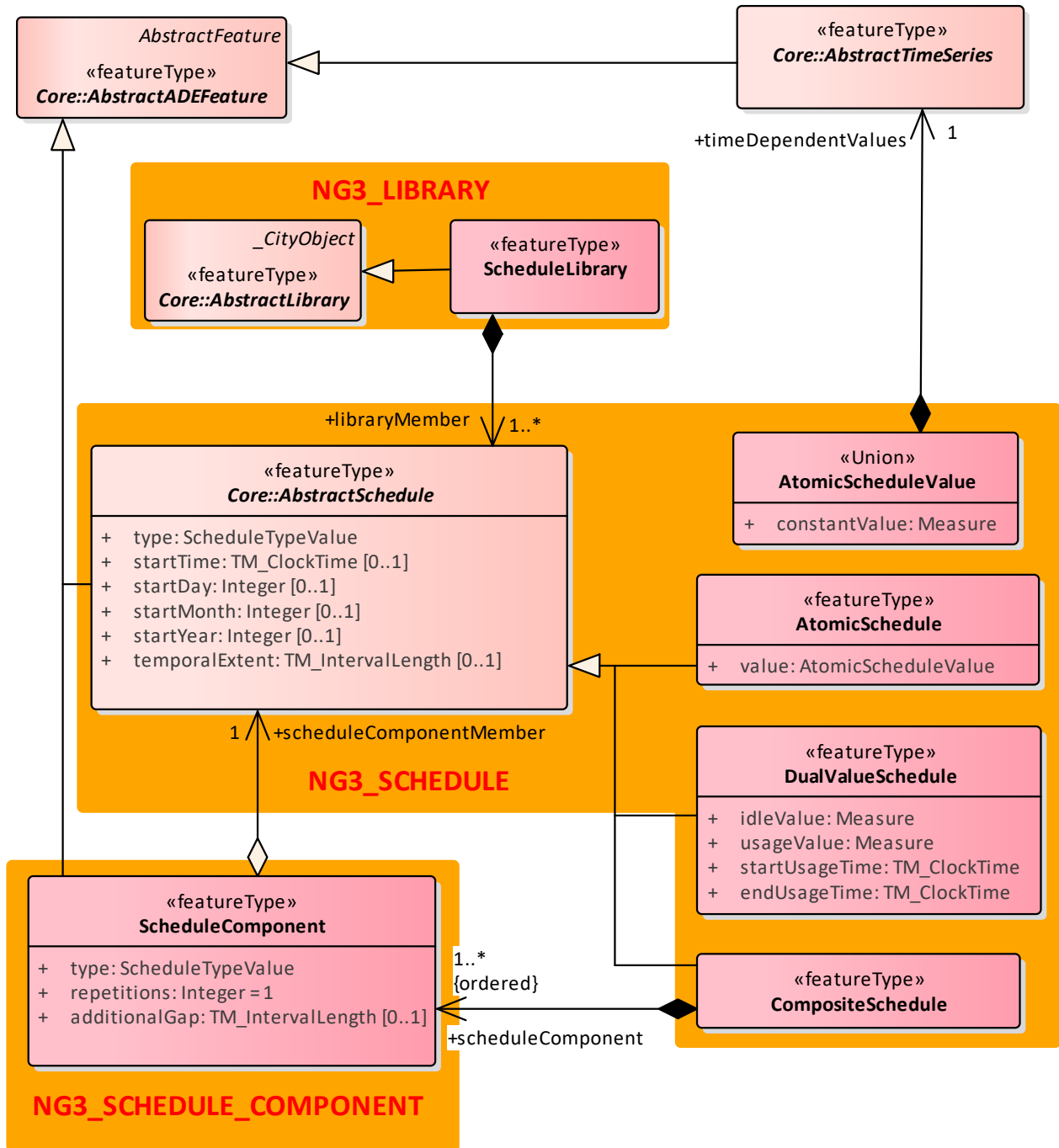
## Resources module



## Layered construction module



## Schedules module

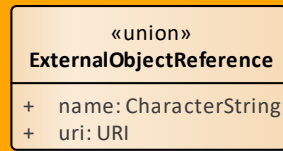
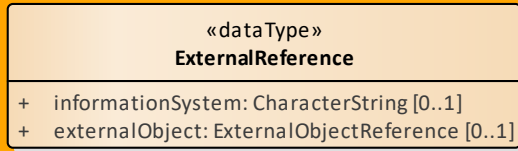


## Time series module

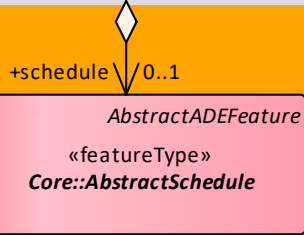
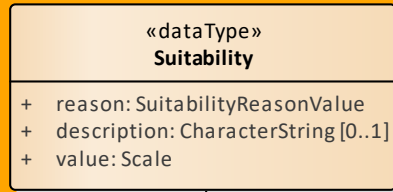


## Supporting classes: DataTypes

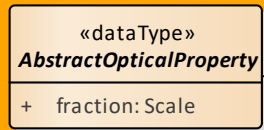
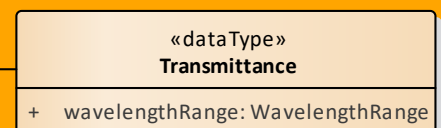
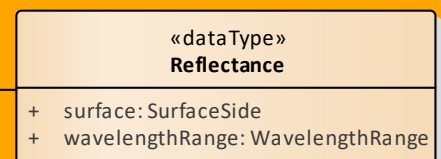
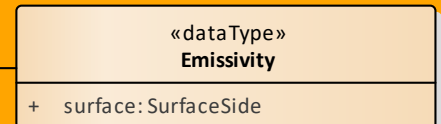
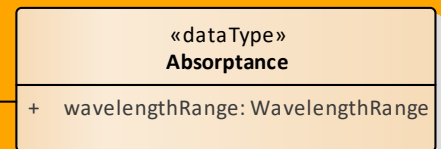
## EXTERNAL\_REFERENCE



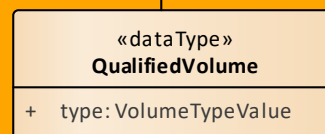
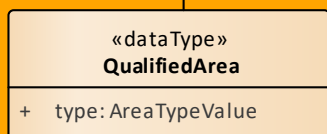
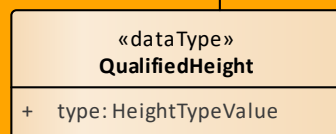
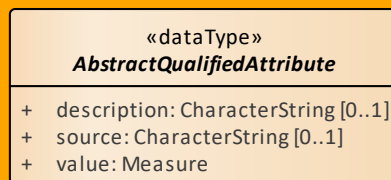
## SUITABILITY



## NG3\_OPTICAL\_PROPERTY



## NG3\_QUALIFIED\_ATTRIBUTE



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